

Measuring Media Bias Through Homepage Duration: Evidence from NYT and WSJ^{*}

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Abstract

We introduce a duration-based measure of newsroom emphasis based on the time articles remain on digital homepages. Using data on more than 100,000 NYT and WSJ articles and 24 million tweets, we examine how reader-engagement proxies and article slant jointly predict homepage survival. For the NYT, more pro-Democrat articles remain longer after conditioning on demand controls. For the WSJ, analogous slant associations are weaker overall and concentrated in selected subsets. The patterns are consistent with editorial weighting that is difficult to reconcile with a purely demand-based interpretation, but they are not definitive causal evidence of supply-side bias. Profit-maximizing outlets may also give greater prominence to content that best retains their core audiences. The results therefore provide a duration-based measurement approach and disciplined descriptive facts for future work on mechanisms.

JEL Classification: D22, D72, D83, L82

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1. Introduction

Digital news creates a new margin along which media bias can operate, not only in what is written, but also in what editors keep prominently displayed and for how long. This paper introduces a novel measurement approach for media bias based on article duration on a newspaper’s digital homepage.¹ Homepage stories occupy privileged digital real estate, and duration on that page captures an outlet’s allocation of attention across articles. We therefore study the determinants of homepage duration while controlling for demand proxies using reader-engagement data. The resulting patterns are informative about when slant and prominence move together after conditioning on demand. We interpret them as evidence consistent with supply-side forces, while recognizing that they do not by themselves rule out all demand-based or audience-mix explanations.

A large literature measures media bias using language, citation patterns, endorsements, coverage choices, or outlet-level slant ([Groseclose and Milyo, 2005](#); [Gentzkow and Shapiro, 2010](#); [Puglisi and Snyder, 2011](#); [Baum and Groeling, 2008](#); [Martin and Yurukoglu, 2017](#)). See also [Puglisi and Snyder \(2015\)](#) and [Gentzkow et al. \(2015\)](#). Those approaches have been highly informative, but they mostly study non-digital or non-prominence-based margins of bias. Our contribution is to study a distinctly digital editorial margin: how long a given article remains on the homepage, conditional on observed demand. By combining text-based slant with homepage duration, we measure within-outlet allocation of attention rather than only average outlet ideology.

This margin matters because homepage placement affects what readers see first, what remains salient, and which stories receive repeated exposure. For that reason, systematic differences in homepage duration can contribute to agenda-setting and may exacerbate polarization or affect democratic discourse, even when article text is unchanged ([Allcott and Gentzkow, 2017](#); [Zhuravskaya et al., 2020](#)). At the same time, our goal is measurement

¹In this paper, we follow [Gentzkow and Shapiro \(2010\)](#) and [Gentzkow et al. \(2015\)](#) to define bias as “systematic differences in the mapping from facts to news reports, that is, differences which tend to sway naive readers to the right or left on political issues.”

rather than a definitive causal decomposition of newsroom motives. Insider accounts such as [Bennet \(2023\)](#) point to both shifting reader demand and newsroom-side changes as possible sources of bias, and our analysis is designed to provide disciplined descriptive evidence on that distinction rather than a final resolution.

We apply this approach to articles featured on the *New York Times* (NYT) and *Wall Street Journal* (WSJ) homepages, the two largest daily newspapers in the United States ([Statista, 2024](#)). These are not just major newspapers, but they represent contrasting political ideologies, with the NYT seen as liberal and the WSJ as conservative. This enables a balanced analysis of ideological influences on journalistic practices. There are other reasons for focusing our analysis on these two newspapers.²

Our study uses a unique dataset that spans August 2021 to May 2023 for the NYT and October 2022 to May 2023 for the WSJ. The dataset includes over 80,000 articles from the NYT and more than 20,000 from the WSJ, augmented by data from Twitter (now X), collected via the Twitter API as a measure of reader engagement. Specifically, our analysis covers approximately 22 million tweets linked to NYT articles (averaging about 275 tweets per article) and nearly 2 million tweets for WSJ articles (averaging about 100 tweets per article). We provide a range of evidence in Section 4.2.1 that tweets accurately measure reader engagement.³

²First, the NYT and WSJ are far and away the most prominent US newspapers, and so understanding the factors driving their decision-making is an important first step to a general theory of media bias. Second, these papers share key characteristics with other papers so the results may generalize: for example we show that the percent of reader visits starting at the homepage is comparable to other national publications. Third, our work complements prior work which has studied local newspaper bias using very different identification strategies ([Gentzkow and Shapiro, 2010](#)).

Focusing on the NYT and WSJ has other advantages. Their extensive presence across multiple platforms enhances our comparative media analysis, offering a contrast to the single channel news-feed for other outlets. In particular, we can and do link the articles appearing on their homepages to their corresponding print editions and podcasts, providing a comprehensive understanding of their editorial strategies, which is not possible with broadcast media like Fox News and CNN.

³We used highly granular data from a representative online panel covering the eight-month period when

To analyze article sentiment and political orientation, we used textual analysis to calculate sentiment scores and pro-Democrat scores. These pro-Democrat scores, which reflect an article’s political slant, were derived using a machine-learning model trained on tweets from approximately 2,000 American politicians labeled with their political affiliations (this measure is strongly correlated with a revealed preference partisan measure based on retweets by a broader set of individuals with known ideological profiles). Summary statistics indicate a liberal slant in NYT articles, with an average pro-Democrat score of 0.6 (where 1 represents full pro-Democrat alignment and 0 indicates complete pro-Republican alignment), while WSJ articles are closer to neutral, with an average score of 0.5.

Our primary analysis employs a parametric survival model to study how long articles stay on newspaper homepages. We show that demand-side factors, especially Twitter activity, play an important role. Articles receiving more tweets per hour consistently show a lower hazard rate, indicating prolonged presence on the homepage.⁴

Even after accounting for demand-side proxies, slant remains associated with homepage presence, particularly for the NYT where more pro-Democrat articles tend to stay longer. In the WSJ, these associations are generally weaker overall, but appear in selected contexts such as U.S. news and politically tagged content. We continue to find related patterns when we focus on articles placed at the very top of the homepage. An additional matched-story comparison that pairs similar stories across newspapers yields qualitatively similar patterns, and the details are reported in Appendix H. Quantitatively, the baseline

we could track both papers. We show that tweets are highly correlated with user page views from this panel at the article, article-day, and article-hour levels (the correlation approaches one at the article level, and the correlations and statistical significance also hold when we look at each paper separately). We also show that tweets are strongly correlated with internal metrics at the NYT and with web traffic (from a second online panel) for the WSJ.

⁴One potential concern is that the feedback effect of homepage placement on tweet counts might contaminate our demand estimate. We provide a range of conceptual and empirical evidence in Section 5.1.2 that this is not an issue.

hazard estimates imply that a one-standard-deviation increase in slant is associated with about a 10% lower NYT exit hazard in the pooled specification (hazard ratio 0.897).

If supply-side forces are important, greater competition should tend to reduce distortions (the opposite comparative static can arise under demand-driven bias models). We compare homepage editorial decisions for platforms in very different competition environments, print-edition and podcasts, in Appendix J. The results are consistent with the supply-side bias model. We treat this cross-format analysis as supplementary and interpret it cautiously, since audiences, editorial teams, and production constraints differ across formats.

Article-level measurement is useful for our question because newspaper partisanship is not monolithic: the NYT has conservative articles and the WSJ has liberal articles. Work on story-level partisan appeal likewise emphasizes meaningful heterogeneity within sources ([Green et al., 2025](#)). In our data, within-newspaper diversity across articles is far larger than the between-newspaper average difference. Some of this heterogeneity likely reflects news flow, since stories can have an underlying partisan orientation. The key margin for our analysis is therefore prominence: which stories are emphasized and for how long.

Finally, an alternative interpretation remains viable and we make it explicit. A profit-maximizing outlet may cater to its niche audience even without ideological intent from ownership or top management. For example, if NYT readers are more responsive to left-leaning frames, longer homepage duration for such stories can arise from demand heterogeneity plus editorial optimization. Our design narrows this concern by controlling for observed engagement, but cannot fully eliminate it.

The remainder of the paper is as follows. Section 2 shows how our work connects to and advances several active research areas, and Section 3 discusses the importance of the homepage to news media websites. We describe the data in Section 4. We then discuss our main analysis and present our empirical results in Section 5. The appendix contains validation, robustness, matched-story, leading-news, and cross-format analyses. The last

section concludes.

2. Literature

Perhaps the central question in this literature is the source of bias. One strand focuses on demand-driven bias: readers prefer news that affirms their own political views (Mullainathan and Shleifer, 2005). Another set of papers focuses on supply-side bias, which stems from the ideological leanings of media owners, editors, or reporters (Gentzkow and Shapiro, 2006). Our paper studies the relative importance of these two channels. This distinction has real-world importance. Consider the role of competition. Under supply-driven models, competition diminishes bias because consumers have more choices and prefer less bias. Conversely, under demand-driven bias, competition can increase bias through a Hotelling mechanism: while a monopoly locates in the political center to maximize readership and profits, duopolists locate symmetrically on opposite sides of the political middle to focus on left- or right-leaning consumer segments.

There is a large and growing literature assessing the extent and sources of media bias. Groseclose and Milyo (2005) are among the first studies to systematically document media bias at a large scale. They find that all but two (*Fox News' Special Report* and the *Washington Times*) news outlets received scores to the left of the average member of Congress, while NYT received scores far to the left of center.⁵ Gentzkow et al. (2015) and Puglisi and Snyder (2015) review the theoretical and empirical literature on media bias, respectively. Many studies have analyzed several drivers of media bias, including demand for media bias (Huang et al., 2022; Gentzkow and Shapiro, 2010; Martin and Yurukoglu, 2017; Puglisi and Snyder, 2011), supply of media bias (Ansolabehere et al., 2006; Baron,

⁵Groseclose and Milyo (2005) focus on newspaper-level bias measures while our analysis uses the newspaper-article level (their measure is based on think-tank citations, which would be hard to apply to the article level since such cites are infrequent and only occur for certain news topics). Garz et al. (2020) consider article-level bias measures, but they manually rate article bias for a small sample of articles (less than a hundred per source).

2006; Cagé et al., 2022; Demsetz and Lehn, 1985; Martin and McCrain, 2019; Strömberg, 2004), professional norms (Baum and Groeling, 2008; Shapiro, 2016), journalists’ ideology (Boxell and Conway, 2022; Cagé et al., 2022), and market competition (Chan and Suen, 2008; Mullainathan and Shleifer, 2005; Gentzkow and Shapiro, 2008; Qin et al., 2018). Several papers also study the historical evolution of partisan news sources (Gentzkow et al., 2006; Hirano and Snyder, 2024) and the importance of fake news and partisan news in a typical American’s news diet (Allen et al., 2020; Muise et al., 2022). While many of the empirical papers above analyze local newspapers, our paper is one of the first to study supply factors of media bias in major national newspapers.

Our contribution differs from established bias metrics in two ways. First, many existing measures are outlet-level averages, such as citation-based or language-based slant scores, while our outcome is article-level prominence over time. This allows us to study within-outlet allocation of attention. Second, prior text-based measures identify ideological position, but do not directly capture editorial weighting. Homepage duration is precisely a weighting margin. We therefore complement, rather than replace, existing slant measures: text identifies orientation, and duration identifies how strongly each orientation is emphasized in curation.

Our paper contributes to the growing literature on the relationship between social media and news. Several papers study the competition for attention between newspapers and social media (de Cornière and Sarvary, 2022; Jeon and Nasr, 2016), the impact of social media on fake news (Allcott and Gentzkow, 2017; Zhuravskaya et al., 2020), and the impact of social media on news production by mainstream media (Cagé et al., 2022; Hatte et al., 2022). Our paper contributes to the literature by using tweets as a proxy for an article’s popularity and showing that demand-side factors play a role in newspapers’ homepage decisions.

This paper also contributes to our understanding of the impact of technology in general and in the media industry. Athey and Luca (2019) documents the use of various statistical tools made possible by digital technology, including A/B testing, in technology

companies. Previous studies have examined the impact of technology and statistical tools in the media industry, including how A/B testing algorithms are changing workflow and headline writing practices ([Hagar and Diakopoulos, 2019](#)) and article performance ([Leung and Strumpf, 2023](#)), the returns to data and informational externalities associated with algorithmic recommendations relative to human curation in online news ([Claussen et al., 2024](#)), the impact of news aggregators on media companies’ revenue and concentration ([Calzada and Gil, 2020](#); [Calzada et al., 2022](#)), how ad-avoidance technologies might increase advertising clutter and reduce content quality ([Anderson and Gans, 2011](#)), and how NYT expansion of its home delivery affects local newspaper circulation ([George and Waldfogel, 2006](#)). Our results shed light on the use and impact of digital technology, specifically the digital homepage, on media companies’ decisions to promote articles and influence media bias.

Lastly, this paper is related to the literature on digital platform bias. Several theoretical papers explore the reasons behind why platforms might bias the rankings of products ([Armstrong and Zhou, 2011](#); [Hagi and Jullien, 2014](#); [Parker et al., 2020](#); [de Cornière and Taylor, 2019](#); [Bourreau and Gaudin, 2022](#)). Some empirical papers attempt to test for platform bias, including demonstrating bias in Amazon’s “frequently bought together” recommendations or Amazon’s buy box ([Chen and Tsai, 2021](#); [Raval, 2023](#)) or measuring bias in search ranking directly ([Aguar et al., 2021](#); [Farronato et al., 2023](#); [Jürgensmeier and Skiera, 2023](#); [Reimers and Waldfogel, 2023](#); [Teng, 2023](#)).

3. The Importance of the Newspaper Homepage

Our main results focus on article homepage duration. In this section we show the homepage shapes what readers view and read. Senior editors understand this and have stated selecting the content of the homepage is one of their highest priorities. Joe Kahn, the NYT’s executive editor, said in an interview:

I think the difference between coming to the Times at any given moment and going to many other places where you can find news on the Internet is we’re constantly

thinking about the hierarchy of the stories that we’re promoting, which is a big part of my day and a big part of my leadership’s team day—the play and the prominence of the different story lines that we have . . . Multiple times a day, we’re debating essentially what the top of the home screen, what the top of the digital feed should look like . . . The ranking of those [articles], the packaging of those, is very much a human curation task and an editing task. And we try to put them together really with a lot of intent every day and multiple times a day ([Malone, 2024](#)).

The homepage is the most critical digital real estate of a newspaper website, acting as the main access point for articles and multimedia content. A 2011 Pew Research Center study highlights its significance, noting that for the majority of sites (21 out of 25) analyzed, including NYT and WSJ, the homepage is the most visited section ([Olmstead et al., 2011](#)).

Recent data further emphasizes the homepage’s crucial role in attracting readers. According to the 2022 annual report from The New York Times Company, their homepage NYTimes.com received approximately 99 million unique monthly visitors from the United States via desktop and mobile devices. This figure increases to about 145 million globally, demonstrating its extensive digital reach across diverse global audiences ([New York Times, 2022](#)).

Most readers start their visit on the newspaper’s homepage, so articles there are more likely to be viewed. Table 1, using data from SimilarWeb Pro between May and June 2023, shows that over 50% of total traffic (desktop and mobile) for each of the NYT and WSJ is direct traffic.⁶ Direct traffic occurs when the user starts reading by opening the newspaper app or enters a URL or bookmark into their browser, which in almost all cases means the user is starting on the paper’s homepage. This is a conservative measure of the homepage’s importance since some of the other traffic may also start on the homepage (this might

⁶Direct traffic is similarly important to other media sources. We collected the top ten comparable media sites to each of the NYT and WSJ, which SimilarWeb provides based on overlap of their audience, content, keywords, and referrals. These lists overlap and include a range of U.S. and non-U.S. print and broadcast sources, such as Fox News, Washington Post, AP, ABC News, the Economist, BBC, the Guardian, and Bloomberg. Using all worldwide traffic, the NYT competitors had an average of 44% direct traffic while the WSJ competitors had an average of 43%.

include referrals, paid or organic search, or social media). Such statistics indicate that a significant majority of visitors are likely accessing the site directly, presumably via the homepage, rather than through secondary channels or external referrals. This confirms the homepage’s crucial role for digital news and emphasizes the importance of editorial decisions about what content to spotlight there.

Table 1: % of Direct Traffic to NYT and WSJ Websites

	NYT		WSJ	
	US	World	US	World
Mobile	48.4%	46.7%	51.5%	49.9%
Desktop	59.8%	57.9%	63.4%	61.3%
Total	52.3%	50.4%	56.3%	54.6%
Data: SimilarWeb Pro (May-June 2023)				

4. Data

We used a scraper program to assemble our article database for the NYT and WSJ, and complemented the data with tweet data from the Twitter API. Additional details on these data are presented in [Leung and Strumpf \(2023\)](#).

4.1. *NYT and WSJ Data*

Our scraper collected data from the NYT homepage, <https://www.nytimes.com>, every minute from August 2021 through June 2023. On each cycle it collected all articles on the NYT homepage, including the headline, URL, and article ID. The scraper also collected information from the official NYT API, <https://developer.nytimes.com/apis>, on every NYT article, including those not on the homepage. The API returned each article’s metadata, including article ID, print headline (if any), when the article was first posted

online, news tone,⁷ section, tags (keywords), and a subset of the article text (abstract and first paragraph) for all articles published anywhere on the digital NYT in the sample period. The API also listed the top 20 articles in terms of email, share, and views for each hour in the sample period. Overall, there were 81,692 articles published by the NYT in the sample period, and 48,501 (59%) of these were put on the homepage at some point.

For the WSJ, we began scraping the WSJ homepage, <https://www.wsj.com/>, on October 8, 2022, using a procedure similar to the NYT scraper. The WSJ scraper collected all articles on the WSJ homepage, including the headline, URL, and article ID. Because the WSJ lacks an API, we scraped the WSJ archive, <https://www.wsj.com/news/archive/years>, to obtain the URLs of all WSJ articles in the sample period and then used a Python package, *newspaper3k*, to extract article metadata, including article ID, when the article was first posted online, section, keywords, and a subset of the article text (abstract and first paragraph). As with the NYT API, the WSJ archive included articles that never reached the homepage. Overall, there were 23,259 articles published by the WSJ in the sample period, and 62% (14,379) of these were put on the homepage at some point.

There are several things worth noting. First, our WSJ homepage scraper was not able to scrape the op-ed articles for technical reasons.⁸ Second, we did not have information for the WSJ on print headline and news tone as we did for the NYT. Instead, we categorized an article’s news tone by its section, which aligns with the NYT sorting.⁹

⁷The three NYT news tones are “News” (which is mainly hard news), “Opinion” (which is mainly op-ed), and “Feature” (which is mainly soft news).

⁸The newspapers have different homepage curation policies for Opinions, so these articles would need to be treated separately in our analysis. We show our results hold section-by-section, so the missing data are not driving our conclusions. We were able to collect op-eds from the WSJ archive, but this source contained no information about the homepage.

⁹The “News” tone includes sections including “Business”, “Economy”, “Markets”, “News”, “Politics”, “Science”, “US”, and “World”. The “Feature” tone includes sections including “Arts”, “Life”, “Style News”, “Style and Substance”, “WSJ Puzzles”, “Tech”, “Real Estate”, and “Magazine”.

4.1.1. Textual Analysis

Following [Leung and Strumpf \(2023\)](#), we used textual analysis to construct two measures for the abstracts, and for tweets in later sections, of an article. The first measure is a sentiment score of the text, which ranges from -1 (most negative) to 1 (most positive). In particular, we used the VADER sentiment analysis tool ([Hutto and Gilbert, 2014](#)) of the Natural Language Toolkit (NLTK) in Python ([Bird et al., 2009](#)) to implement sentiment analysis for the abstracts of each article in the sample. As [Hutto and Gilbert \(2014\)](#) shows, VADER’s model incorporates syntax and punctuation rules, and is validated with human coding, making its sentence prediction 55–96% accurate, which is on par with Stanford Sentiment Treebank, a method that incorporates a more complex computational algorithm. VADER also includes the word banks of established tools like LIWC, ANEW, and GI, as well as special characters such as emoticons and cultural acronyms, for example LOL, which makes it particularly suited for social media jargon.¹⁰ In addition to social media text, [Hutto and Gilbert \(2014\)](#) also shows that VADER performs well across several domain contexts including NYT editorials, movie reviews, and product reviews.

The second measure captures political slant. We followed [Boxell and Conway \(2022\)](#) and [Leung and Strumpf \(2023\)](#) in adopting a machine learning approach to construct our measure of political slant. To construct our training and testing data, we first obtained the Twitter accounts of approximately 2,100 politicians with known affiliations with either the Democratic Party or the Republican Party from ProPublica, an investigative journalism website.¹¹ We then used the Twitter API to obtain all their tweets, approximately 2 million tweets in total.

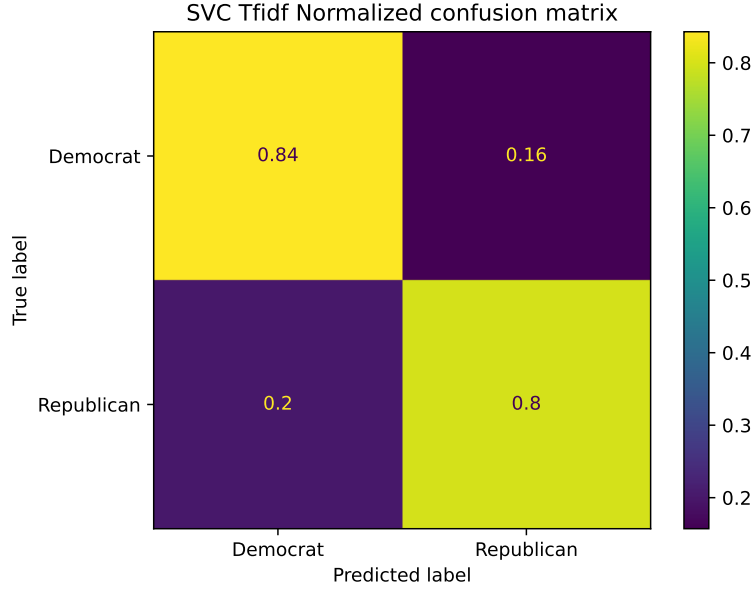
After applying the standard preprocessing steps, we used the Linear Support Vector Classification (Linear SVC) model in the scikit-learn package, a Python module, as our machine learning model.¹² We fit the model using the training dataset, approximately

¹⁰We also used VADER to obtain sentiment scores for tweets, as discussed in Section 4.2.

¹¹We dropped the independent politicians in our sample.

¹²We analyzed both unigrams and bigrams in our machine learning model.

Figure 1: Confusion Matrix of Machine Learning Prediction



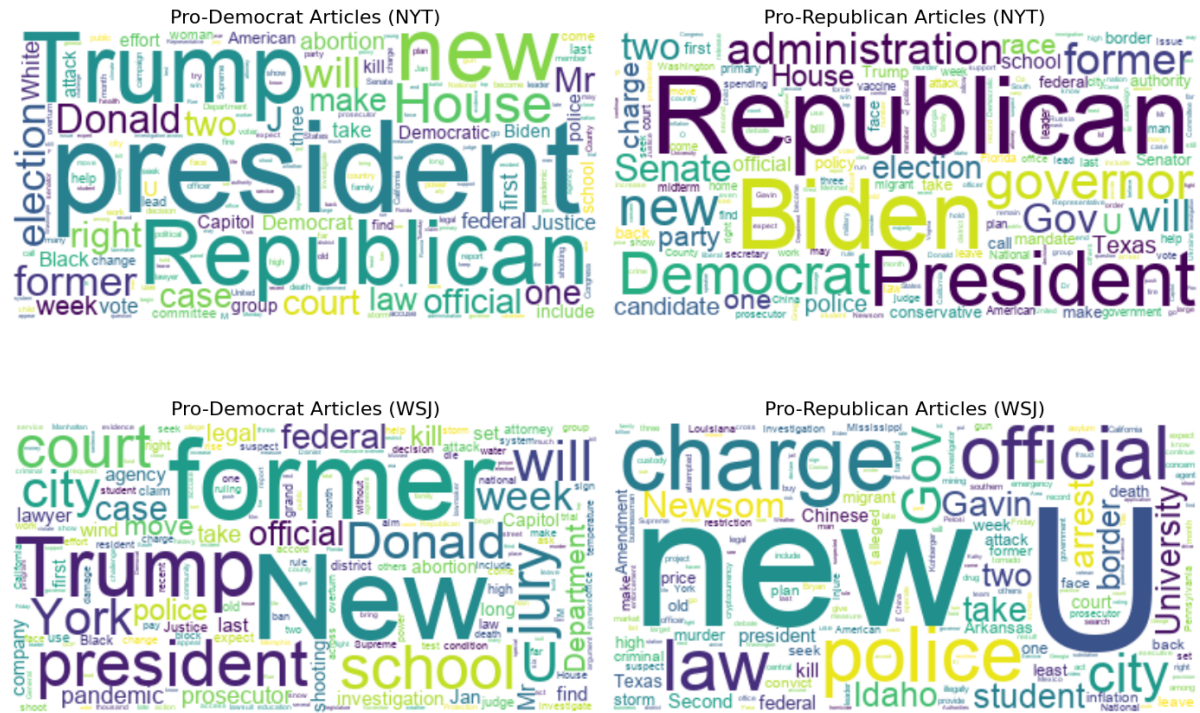
1.4 million tweets. Overall, the model achieved 82% accuracy on the held-out validation dataset of 0.6 million tweets. We also constructed the confusion matrix to evaluate the model fit. It shows the percentage of correct and incorrect classifications based on each class, Democrat or Republican. Figure 1 shows that our model can predict a Democrat’s tweet correctly 84% of the time, with only a 16% chance of being wrongly labeled as a Republican’s tweet. It can also predict a Republican’s tweet correctly 80% of the time, with a 20% chance of being wrongly labeled as a Democrat’s tweet.

We apply the model to “predict” whether the text in our main dataset is written by a Democratic politician (we presume Democrats are liberal and Republicans conservative). The prediction probability that the text was written by a Democrat, which ranges from 0 to 1, is our measure of slant.

Figure 2 presents lexical patterns within U.S. news articles from the NYT and WSJ.¹³ The word clouds are divided into two main panels: the upper panel displays text clusters for NYT articles, with pro-Democrat articles (pro-Democrat scores above 0.7) on the left and pro-Republican articles (pro-Democrat scores below 0.3) on the right. The lower

¹³We present more word clouds in other news sections in Appendix A.

Figure 2: Word Clouds of US News Articles' Abstracts



panel shows the most frequent terms within WSJ articles, following the same political score categories.

Both the NYT and WSJ pro-Democrat articles exhibit a notable overlap in frequently used terms. Common lexemes such as “former”, “President”, “Trump”, and “abortion” underscore this shared thematic emphasis.

The pro-Republican articles have more varied vocabulary across the two publications. In the NYT, pro-Republican articles predominantly feature terms such as “Republican”, “President”, “Biden”, and “Democrat”. In contrast, WSJ’s pro-Republican word choices include “police”, “arrest”, “Newsom”, and “migrant”.¹⁴ These differences hint at diverging editorial foci and narrative frameworks adopted by the two newspapers.

¹⁴There are a few terms such as “Texas” and “border” which feature in both NYT and WSJ’s pro-Republican articles.

4.1.2. Summary Statistics

Next, we summarize the newspaper data, starting with statistics on the homepage, which is central to our analysis.

Figure 3 displays the homepage duration distributions for NYT and WSJ articles, with the left panel for NYT and the right for WSJ. Notably, 5% of NYT articles and 8% of WSJ articles were removed from the homepage within the first two hours of posting.

Op-ed articles in NYT remain on the homepage longer than other categories, with over 15% staying visible for more than 48 hours, compared to less than 3% for other news tones in both NYT and WSJ.

There are spikes at certain durations showing that both NYT and WSJ editors frequently withdraw articles at specific intervals, such as after 6 or 12 hours, particularly for NYT's Feature articles. This pattern indicates a strategic approach to regularly refreshing homepage content, which influences the exposure duration of various news stories.

Figure 3: Histogram of Homepage Duration by News Tone

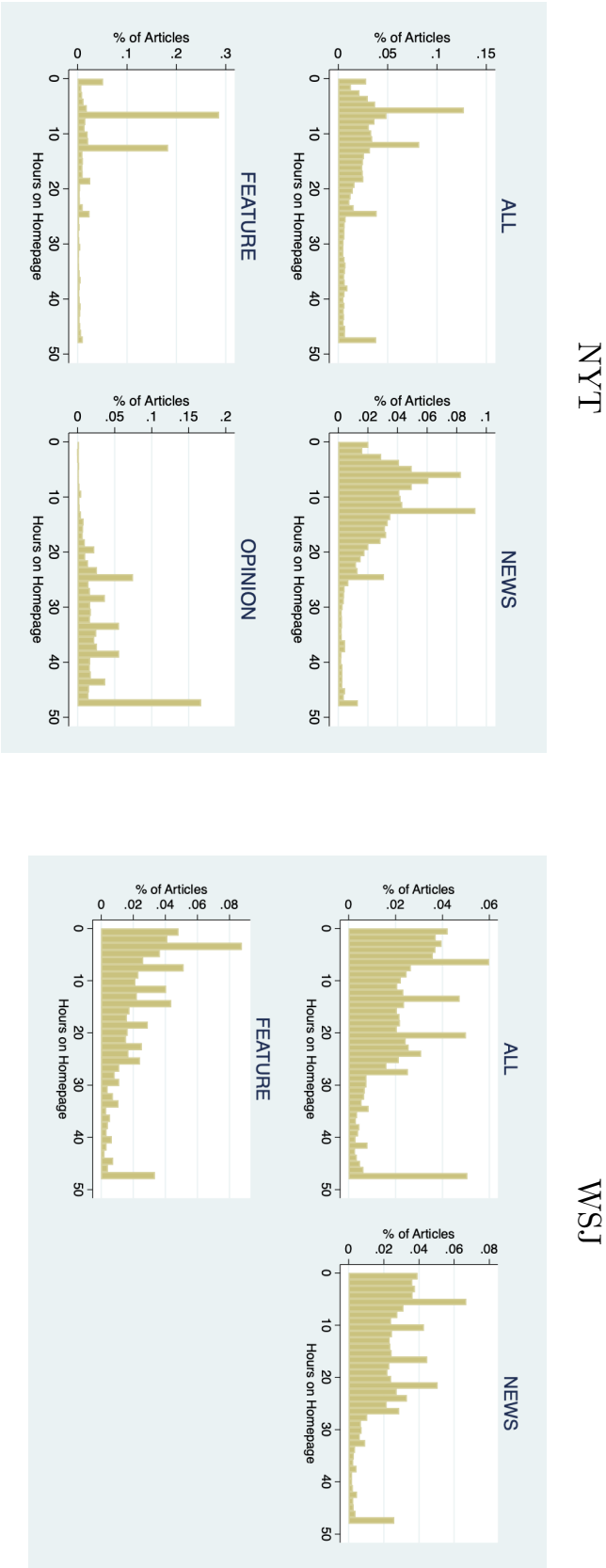


Table 2: Summary of News Articles

	NYT		WSJ	
	Homepage (No)	Homepage (Yes)	Homepage (No)	Homepage (Yes)
Sentiment Scores (Abstract)	0.0690 (0.392)	-0.0140 (0.454)	0.0348 (0.387)	0.0161 (0.423)
Pro-Dem. Scores (Abstract)	0.603 (0.205)	0.598 (0.215)	0.519 (0.213)	0.539 (0.209)
tone (News)	0.517 (0.500)	0.614 (0.487)	0.193 (0.395)	0.643 (0.479)
tone (Opinion)	0.00925 (0.0957)	0.133 (0.340)		
tone (Feature)	0.471 (0.499)	0.253 (0.435)	0.250 (0.433)	0.231 (0.422)
N	33191	48501	6357	14250

Mean coefficients; SD in parentheses

Table 2 presents summary statistics for articles in the two newspapers, including the political slant of their content, and the sentiments conveyed in their headlines and abstracts.¹⁵

First, 60% of NYT articles and 70% of WSJ articles are featured on their homepages at some point. Since most articles reach the homepage, we do not focus on the entry decision in the main analysis.

Second, the political slant scores show that NYT articles are generally more pro-Democrat or liberal, with average scores around 0.6, while WSJ articles display more moderate scores a bit above 0.5. These mean values corroborate the commonly perceived political inclinations of these two major publications. However, note the large dispersion (standard deviation) around these means: each newspaper contains articles with a wide range of partisan slant.

¹⁵The WSJ tone rows do not sum to one due to our data collection procedure. The first column includes articles from the WSJ archives, including Opinion and articles that do not follow the NYT tone categories, such as C-Suite articles. The second column is from our homepage scraper, which omitted the former as discussed in the text. The rows sum to 0.87. Only the second column is relevant for our estimates in the results section, which are based on homepage articles.

Last, there are limited differences between articles that do or do not reach the homepage. Most important for our analysis, homepage articles have a similar political slant as other articles at each newspaper. This is especially true for the NYT, and for the WSJ homepage articles are slightly more liberal than those not featured there. In terms of sentiment, articles on the homepage exhibit slightly more negative tones than non-homepage articles for both newspapers.

4.2. *Twitter API*

Ideally, we would assess article popularity using actual view and share data from the NYT and WSJ, but only limited analytics data are available. Instead, we approximate reader demand by counting the number of tweets linking to an article’s URL. We provide empirical evidence below that tweets closely match various measures of reader engagement.

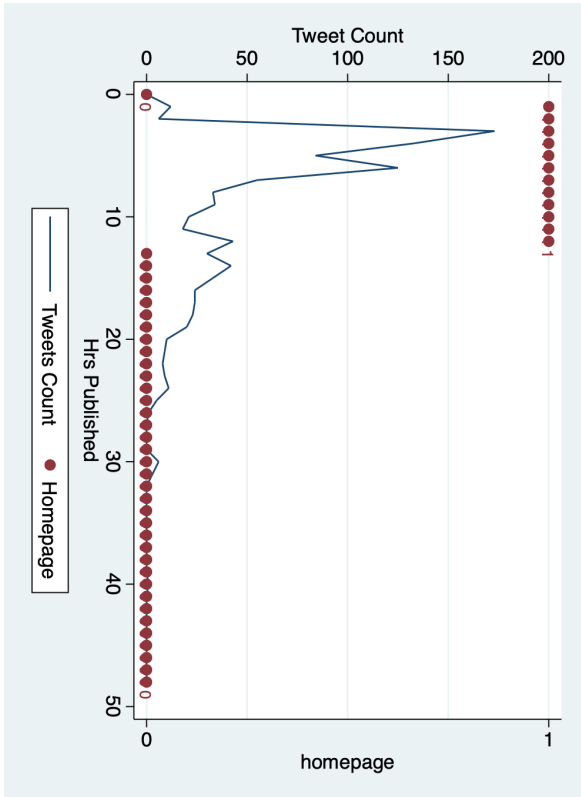
Twitter was a significant news source during our sample period, according to a 2021 Pew Research Center survey of 2,548 users ([Anderson et al., 2021](#)). The survey reported that 23% of Americans used Twitter and about 69% of U.S. Twitter users obtained news from the platform. Despite general skepticism toward social media information, only 27% of Americans trusted it. Among Twitter news consumers, however, two-thirds expressed some level of trust in the news they obtained from Twitter. Reliance on Twitter for news had grown: 70% of Twitter news consumers followed live events on the site, an increase from 59% in 2015.

Between January 2021 and June 2023, Twitter made its entire tweet archive accessible to academic researchers. Using the Twitter API, we extracted all tweets containing URL links to NYT and WSJ articles in our sample. For each tweet, we gathered content, date and time, likes, retweets, and author information such as name and Twitter ID. We also calculated sentiment scores for each tweet similarly to the article abstracts. Our database includes over 24 million tweets: 22 million for the NYT sample (approximately 275 tweets per article) and 2 million for the WSJ sample (approximately 100 tweets per article).

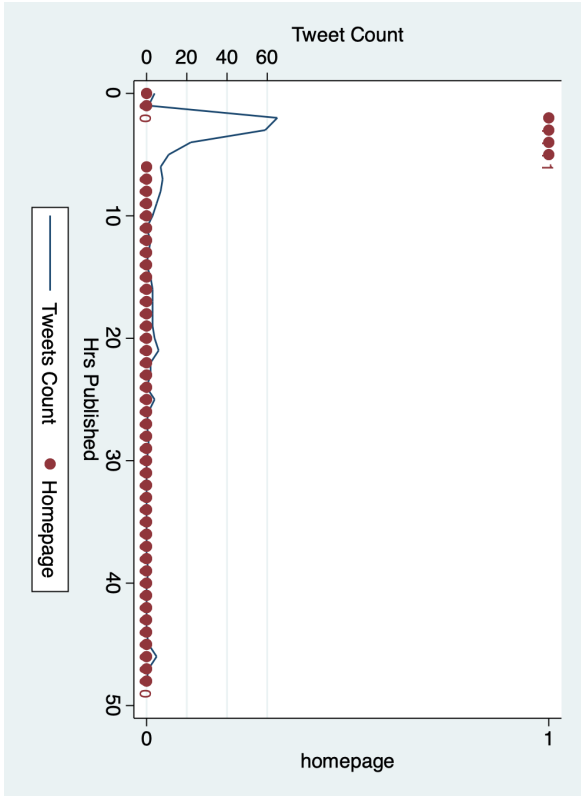
Figure 4 illustrates the interaction between social media activity (measured in tweet

counts) and an article's presence on a newspaper's homepage. It compares two articles covering the same-sex marriage vote: NYT's "Here Are the 12 Republicans Who Voted to Protect Same-Sex Marriage" and WSJ's "Senate Votes to Advance Same-Sex Marriage Legislation."

Figure 4: A Tale of Two Same-Sex Marriage Articles



“Here Are the 12 Republicans Who Voted to Protect Same-Sex Marriage” (NYT)



“Senate Votes to Advance Same-Sex Marriage Legislation” (WSJ)

Both articles had a liberal-leaning tone, with pro-Democrat scores of 0.62 for the NYT and 0.71 for the WSJ. On social media, the NYT article garnered 533 tweets in the first six hours, peaking at nearly 180 tweets in the fourth hour, while the WSJ article attracted 169 tweets, peaking at about 60 tweets in the fourth hour. Adjusting for typical engagement levels, NYT articles generally receive three times more tweets, so the engagement for both articles was comparable.

However, the homepage duration differed significantly: the NYT article remained on the homepage for 12 hours, while the WSJ article was featured for only 4 hours.

These contrasting trajectories suggest that article slant and editorial prominence may co-move in ways not fully captured by contemporaneous demand proxies. While demand-side factors like social media activity clearly play a role, they do not fully explain homepage duration in this example. This helps motivate our more formal investigation of slant-duration correlations in editorial decisions.

4.2.1. Validation of Tweets as Measure of Reader Engagement

In this subsection we establish that tweets are an accurate measure of reader engagement with articles (neither the NYT nor the WSJ allows researchers to access internal article-level analytics).¹⁶ We consider three data sources, each of which brings unique insights, and the suite of evidence builds a strong case for our measure.

The first is the most direct test using page views of individual articles from a large panel of internet users. These data come from Luth Research which has been conducting consumer surveys for about fifty years. Their consumer panel includes over three million individuals. The panelists are linked to a confirmed physical address, and joining the panel requires double opt-in. Of this panel, about 80,000 have opted into being digitally tracked via Luth’s proprietary software, ZQ Intelligence. There is 24/7 tracking of panelists’ internet browsing within websites and apps, including PCs, phones, and tablets. We used their U.S. panel, which has representative demographics of the U.S. online population

¹⁶Section 5.1.2 addresses another potential challenge to the tweet-based measure of demand.

in terms of gender, race, and age distribution. Luth aims to provide a representative sample of internet browsing behavior, comparable in data and methodology to comScore, Nielsen/NetRatings, and SimilarWeb, which we discuss below.

We considered the period for which we had data for both newspapers, roughly October 2022 through May 2023. This covered about 50,000 individual articles, about half from each of the NYT and WSJ. For this period, Luth provided us with a list of every interaction each of their U.S. panelists had with any webpage from the NYT or WSJ domain, including browsers or apps in the case of mobile users. Each interaction included the URL and timestamp. This allowed us to construct a granular measure of page views for each URL in our sample. We omitted visits to URLs not in our sample, which were primarily older articles. From these joined data, we compared page views and tweets.

Table 3 shows various correlations of our tweet count measure with page views from the Luth panel (we present separate values for the full sample as well as for each newspaper). In all cases there is a large, positive, and statistically significant correlation (we use the standard two-sided t-statistic to test the null of no correlation). The top panel aggregates the results to the article level: the total number of tweets and page views ever observed. We find values around +0.75, not perfect correlation but relatively close (the values for each newspaper are comparable). The second panel restricts values to tweets/page views over the first twenty-four hours after each article is published online, which is the time interval for much of our analysis. The correlations are only slightly smaller, roughly +0.66, though a bit less for the WSJ. The next panel shows the correlation at the article-day level, providing some time granularity. The values remain large, positive, and statistically significant, with values exceeding +0.50. The final panel looks at the article-hour level, matching the time granularity of the main results. The correlations here are about +0.33 (less for the WSJ) and statistically significant.

In total this is strong support for our use of tweets as a measure of reader engagement. The values here understate the connection between tweets and page views, since there are two factors reducing the correlations. The Luth panel is a sample, and it only includes US

users, so the population correlations are likely much higher (the article-hour correlations are noticeably weaker in early AM ET hours when readership is likely skewed to European readers as US readers are sleeping).

Table 3: Correlation of Tweet Count and Luth Panel Page Views

	Full Sample	NYT Articles	WSJ Articles
Article-level	0.772 (0.000) [N=47876]	0.766 (0.000) [N=27266]	0.849 (0.0000) [N=20610]
Article-level, through twenty-four hrs after publication	0.661 (0.000) [N=47876]	0.672 (0.000) [N=27266]	0.588 (0.00) [N=20610]
Article-Day-level	0.512 (0.000) [N=164241]	0.520 (0.000) [N=81801]	0.495 (0.000) [N=82440]
Article-Hour-level	0.334 (0.000) [N=2840711]	0.349 (0.000) [N=1336181]	0.241 (0.000) [N=1504530]

Correlation coefficients; significance-level (based on t-statistic) in parentheses, and sample size in square brackets.

Data are for October 2022 through May 2023.

Tweet data from main sample, and Page Views are from the Luth Panel discussed in the text.

The second piece of evidence is coarser but comes from actual NYT analytics for several measures of reader engagement (these values come from the NYT API). Table 4 shows the average hourly tweet count for articles in the top 20 positions in terms of reader views, shares, and emails (these data cover August 2021 through June 2023, the period for which we have NYT data). There are significant tweet differences across tiers of demand-side article popularity.

Articles below the top 20 have 4.5 to 5 tweets per hour, while top 20 articles average between 12 and 24 tweets per hour. Within the top 20, top 10 articles have even higher tweet counts than those ranked 11–20. This pattern is most evident in share rankings, with top 10 articles averaging 24 tweets per hour versus 16 for the 11–20 group. A similar trend, though less pronounced, is seen in email rankings, with top 10 articles averaging 17 tweets per hour compared to 12 for those ranked 11–20.

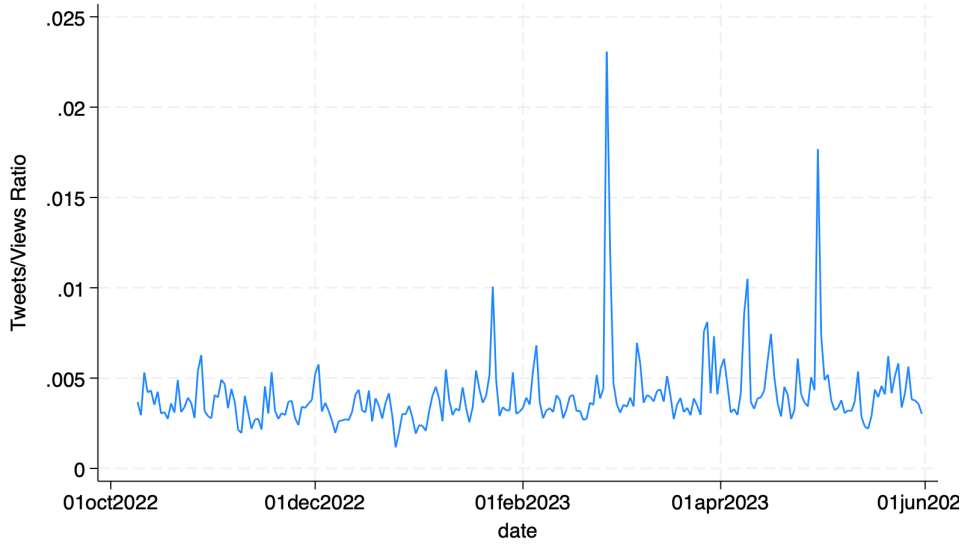
The final piece of evidence is for the WSJ. Since we did not have internal analytics,

Table 4: NYT Articles' Rank and Hourly Tweets Count

	Views	Share	Email
Outside Top 20	4.89 (46.66)	4.54 (40.90)	5.01 (45.53)
Rank 11–20	13.20 (71.65)	16.03 (114.80)	12.33 (92.24)
Rank Top 10	14.96 (66.20)	24.32 (48.25)	17.01 (87.11)

Means; standard deviations in parentheses.

Figure 5: Daily Tweets/(Similar Web) Views Ratio for WSJ (Oct 2022 to May 2023)



we used data from SimilarWeb, a web analytics company that measures internet traffic based on its own internet panel as well as partnerships with certain sites. From this site we obtained daily views of the WSJ homepage for our sample period, October 2022 to May 2023, and compared them to daily aggregated WSJ tweets. The correlation between tweets and views is notably high at 0.65, indicating a strong relationship between the two measures. Figure 5 further illustrates that the tweets-to-views ratio remains fairly constant at approximately 0.004, with only a few outlier days. This is a helpful complement to our Luth data since it includes all traffic to the paper and is based on an alternate sample that also includes non-U.S. users.

Overall, these three datasets help validate the use of tweet counts as a reliable measure of an article’s popularity, supporting its application in our analysis.

4.2.2. Validation of Article Slant Measure

Our measure of article partisanship is based on machine learning of article text. To help validate this approach, we considered an alternative measure based on the behavior of Twitter users. The idea is that if an article is being favorably propagated by mainly right [left]-leaning accounts, then this suggests the article has a right [left] slant. We checked whether this alternative approach correlated with our textual measure.

We examined the retweeting pattern of a large set of Twitter users of known ideology. Article retweets are considered endorsements, so the alternative measure is based on the ideological composition of the retweeting accounts. Through special arrangement with the site Ground News, we obtained access to their BlindSpotter database of 3.4 million Twitter accounts, with each account’s ideology score based on an average of 70 news engagements. We used these data to measure the ideological distribution of retweets of each article in our data.

Our main measure and this revealed-preference measure of article slant are strongly correlated: for both the NYT and WSJ, the alternative measure explains between 70 and 90% of the variation in the machine learning partisan measure. Background on the alternative measure, including key assumptions, as well as empirical estimates are presented in Appendix B.

4.2.3. Summary Statistics and Homepage’s Causal Impact on Tweets

Table 5 summarizes Twitter engagement for articles from the NYT and WSJ, aggregating activity across each article’s lifetime.

Table 5: Summary of News Articles’ Tweets Count

	NYT		WSJ	
	Homepage (No)	Homepage (Yes)	Homepage (No)	Homepage (Yes)
Tweets Count	69.0 (559.9)	409.6 (1510.7)	33.6 (193.3)	138.3 (1028.2)
Tweets % (Neg)	0.0581 (0.0890)	0.0778 (0.0788)	0.0441 (0.0800)	0.0859 (0.0889)
Tweets % (Pos)	0.0832 (0.0993)	0.0833 (0.0680)	0.0602 (0.0913)	0.0957 (0.0821)
N	33191	48501	8880	14379

Means; standard deviations in parentheses

First, the data show that NYT articles are more than twice as popular on Twitter as WSJ articles, with average tweet counts of 277 and 98, respectively. For articles featured on their respective homepages, the gap widens: NYT articles receive about three times as many tweets (410 on average) as WSJ articles (138 on average).

Second, homepage articles attract significantly more tweets for both publications. NYT articles on the homepage garner about six times more tweets, while WSJ articles see a fourfold increase. This suggests that homepage placement either amplifies an article’s visibility and popularity or that more newsworthy articles are featured on the homepage, or both. Appendix C presents estimates of the causal impact of homepage presence on tweet count, using time-of-day homepage updating patterns as an instrumental variable.¹⁷ There are distinctive hours of the day in which large purges of homepage content occur, likely associated with editor shift changes. These hours occur throughout the day and so are unrelated to article newsworthiness, though they do shape which articles are on the homepage. After instrumenting, we find that tweet counts for articles increased significantly when featured on the homepage (35% increase for NYT articles and 162% for WSJ

¹⁷A separate concern is that there are time-of-day patterns to social media posting. But we show in [Leung and Strumpf \(2023\)](#) that tweets mentioning newspaper articles come from users in a wide geographic range over many time zones, and so any diurnal posting patterns would be attenuated.

articles).

Third, 14–18% of tweets for articles from both newspapers express a discernible sentiment, either positive or negative, with a slight preference for positive sentiment. This indicates a relatively balanced sentiment distribution in reader responses.¹⁸

5. Homepage Survival Analysis: Model and Results

Our main analysis focuses on article homepage duration. The previous sections have shown that homepage placement raises reader engagement, and editors devote substantial effort to curation. Extended homepage duration is therefore a revealed measure of editorial emphasis. In this section we estimate how duration covaries with demand proxies and article slant.¹⁹

5.1. *Model and Identification*

5.1.1. Model

We estimate a survival model to understand the distribution of a particular “time to event”, which in our case is the duration an article stays on the homepage of the newspaper’s website.²⁰ In the analysis we include both supply- and demand-side factors, variables

¹⁸Following [Hutto and Gilbert \(2014\)](#) we score each tweet based on the proportion of text which receives a positive, negative, or neutral score. The percent values in the table are the average value for the positive and negative values across all tweets. Note that the overall tweet sentiment is a compound score based on the summed valence scores of each word in the lexicon.

¹⁹We do not focus on the entry decision, since most articles reach the homepage at some point. We also showed earlier there is little difference in political slant between articles that do and do not get on the homepage.

²⁰Approximately 10% of articles were temporarily removed from the homepage and later reinstated, typically after a gap of one to two hours. These occurrences, though infrequent, are included in our main estimates, which track both the initial and subsequent exits. To ensure robustness, we conducted separate estimates that considered either the first exit or the final exit for these reappearing articles. The results remained consistent with those reported in the text, likely due to the limited number of such cases within the overall sample.

such as sentiment, political slant, and tweet counts, as well as other factors that may influence homepage longevity, such as the timing of other new articles (more newsworthy articles may crowd out those currently on the homepage).

The model employed is a parametric survival model with a Weibull distribution. The Weibull model is chosen for its flexibility. It can take on a variety of shapes and thus accommodate a variety of underlying hazard rate patterns. Its hazard function, defined as the risk of removal at any given time, is expressed as:

$$h(\tau; X_{it}, \beta, \lambda, \alpha) = \alpha \lambda (\tau \lambda)^{\alpha-1} \exp(\beta X_{it}), \quad (1)$$

where τ is the number of hours since publication, λ is a scale parameter, and $\alpha > 0$ is the shape parameter. For $\alpha > 1$, the hazard function is monotonically increasing, meaning the risk of removal grows over time. The variable X_{it} is a set of (potentially) time-varying controls for article i at time t , including the natural log of zero-inflated tweets at $t - 1$,²¹ sentiment and pro-Democrat scores of the article, and the number of new articles published by the newspaper at time t .²² Additionally, the model incorporates several fixed-effect dummies for the time of the article’s publication (hour of the day, day of the week, and year-month).

In this study, we are interested in separating observed demand correlates from residual slant-duration correlations, while recognizing that this is not a complete causal decomposition.

On the *demand side*, we capture the article’s popularity through the inclusion of tweet counts as a control. We posit that articles generating higher tweet counts, signaling greater reader engagement and interest, would have an extended presence on the homepage. Sections 4.2.1-4.2.3 provide evidence that tweets capture reader demand.

On the *supply side*, we measure an article’s political slant using pro-Democrat scores

²¹Zero-inflated tweet counts are the number of tweet counts plus one.

²²We use the number of new articles of the same news tone when we restrict the sample to a certain news tone.

derived from the article text. While the NYT is generally more liberal and WSJ more conservative, individual articles within each newspaper vary widely in their political slant. By incorporating pro-Democrat scores, we can determine if articles with a particular leaning remain on the homepage longer than articles with a different leaning but the same demand (tweet counts). If editorial choices beyond observed demand proxies matter, articles that align more closely with the newspaper’s political stance may have extended homepage durations. Key to our approach is the simultaneous control for demand using tweet counts.

Therefore, the inclusion of both sets of variables allows us to characterize how far demand proxies account for duration and what slant-related residual patterns remain.

5.1.2. Challenge to Identification

The key concern is a feedback loop: homepage placement itself can increase tweets (Appendix C), so tweets may partly embed editorial treatment. Our strategy is to ask whether this mechanically contaminates the slant estimates.

We address this in three steps. First, we use an alternative demand proxy with lower direct amplification (retweets) and obtain the same qualitative patterns.

Second, we test the feedback concern directly using the Luth pageviews panel from Section 4.2.1. If feedback dominated tweets, tweet-pageview correspondence should weaken while an article is on the homepage.²³ Instead, it increases: for on-homepage article-time, the overall correlation is 0.389 (0.334 in the full sample), with 0.406 for NYT (0.349 full sample) and 0.281 for WSJ (0.241 full sample). The correlation of demand with tweets is *stronger* when articles are on the homepage, which seems inconsistent with the feedback hypothesis.

Third, under a transparent assumption, any common feedback term factors out of

²³When the article is not on the homepage there is no possible feedback, and so tweets should accurately gauge demand. This would not be true while the article is on the homepage, so tweets would be a noisier measure of demand.

the hazard likelihood. In our risk set, all articles are on the homepage by construction, so a common additive feedback component in tweets does not create differential survival comparisons.²⁴

We can formally show this. Consider the true nonparametric proportional hazard model (of which the Weibull is a special case with a parametric baseline $h_0(t)$),

$$h(t; X) = h_0(t) \exp(X^- \beta^- + X^D \beta^D) \quad (2)$$

where we split the covariates into demand factors X^D and all others X^- . Our proxy for demand is tweets, which (by assumption) includes a time-invariant feedback effect F common to all homepage articles, $X^{Tweets} = X^D + F$. Substituting and simplifying, we have the hazard we estimate (which includes feedback),

$$h^{estimated}(t; X) = \exp(F \beta^D) \times h(t; X) \quad (3)$$

The partial likelihood for the estimated hazard is identical to the one for the true hazard: the $\exp(F \beta^D)$ term drops out of the partial likelihood since the contribution of each term i is,

$$L_i^{estimated}(\beta) \equiv \frac{h^{estimated}(t; X_i)}{\sum_{k \in R_i} h^{estimated}(t; X_k)} = \frac{h(t; X_i)}{\sum_{k \in R_i} h(t; X_k)} \equiv L_i(\beta) \quad (4)$$

where R_i is the risk set of articles which have a duration on the homepage at least as long as i and duration ties are omitted to simplify the exposition. This means the β parameters are estimated without bias even if there is a feedback effect.

Finally, any bias, if it exists, would primarily be limited to the demand term. In particular, even if feedback affects the estimates on the tweets (demand) parameter, it should not bias our estimates on the supply-side variable (political slant). This is because

²⁴One way to test the constant-feedback assumption is to check whether feedback varies with observables.

In Appendix D, restricting to on-homepage article-time, we find no statistically significant differential feedback by article partisanship.

the latter is identified largely from variation in the article text, which is unaffected by feedback. Further mitigating this concern is the rich set of covariates such as publication time, section (news tone), and so on.

5.2. Main Results

In Table 6, we present our survival analysis estimates of (1) separately for the NYT (left panel) and WSJ (right panel). There are results for each paper overall as well as by news tone. The parameter estimates are shown as hazard ratios, which measure the effect of each covariate on the likelihood of an article being removed from the homepage. A hazard ratio less than one indicates that as the covariate increases, the article is less likely to be removed, meaning it stays on the homepage longer.

Table 6: Survival Analysis of Homepage (Hazard Ratios)

	NYT				WSJ		
	News	Opinion	Feature	All	News	Feature	All
Ln Tweets	0.970*** (0.00411)	0.734*** (0.0114)	1.121*** (0.0106)	0.982*** (0.00367)	0.929*** (0.00963)	0.845*** (0.0175)	0.908*** (0.00823)
Sentiment Scores (Abstract)	0.950*** (0.0111)	0.935** (0.0310)	0.997 (0.0205)	0.956*** (0.00928)	0.872*** (0.0201)	0.974 (0.0385)	0.904*** (0.0175)
Pro-Dem. Scores (Abstract)	0.911*** (0.0225)	0.888* (0.0538)	0.874*** (0.0388)	0.897*** (0.0182)	0.986 (0.0472)	0.922 (0.0779)	0.948 (0.0386)
# of New Articles	1.042*** (0.00198)	1.254*** (0.00912)	1.015*** (0.00339)	1.023*** (0.00103)	1.072*** (0.00436)	1.044*** (0.00779)	1.041*** (0.00208)
Observations	384488	223499	137890	745877	141491	46784	198575
Hour FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Day FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes

p-stats are for hazard ratio differences from one.

Estimates for α , λ from (1) are omitted.

Exponentiated coefficients; Standard errors in parentheses

* $p < .1$, ** $p < .05$, *** $p < .01$

A key finding from our analysis is the significant impact of demand-side factors. There is a strong negative relationship between tweet counts at $t - 1$ (a proxy for article popularity) and the hazard rate (the likelihood of an article being removed from the homepage). For the NYT, a one-standard-deviation increase in log tweets (1.37) leads to a 2.5% de-

crease in the hazard rate. This effect is even more pronounced for the WSJ, where a one-standard-deviation increase in log tweets (1.09) leads to a 10.0% decrease in the hazard rate. This indicates that more popular articles, as measured by tweet counts, tend to remain on the homepage longer, underscoring the role of reader demand in determining homepage duration. This result also holds for different news tones and is robust to a variety of changes.²⁵

Slant-duration correlations are strongest for the NYT. A one-standard-deviation increase in an article’s pro-Democrat score is associated with a 2% decrease in the hazard rate, which translates into approximately an additional 30 minutes of homepage exposure. In the pooled NYT specification, the hazard ratio on pro-Democrat score is 0.897, implying roughly a 10% lower exit hazard per unit increase of the score. These magnitudes are economically meaningful but should be interpreted as conditional associations, not definitive structural effects.²⁶

For the WSJ, the slant-duration relationship is generally weaker and often not statistically distinguishable from zero in pooled specifications. In Table 6, the all-article hazard ratio on pro-Democrat score is 0.948 with standard error 0.0386, implying an approximate 95% confidence interval of about [0.87, 1.02]. This interval rules out very large effects in the pooled sample, but still allows modest effects of either sign. At the same time, section- and tag-level estimates indicate selected contexts (notably U.S. and political tags) where

²⁵It is possible some tweets criticize an article rather than reflecting demand to read it. To address this potential concern, we replace the total number of tweets with the count of positive sentiment tweets only. Results reported in Appendix E remain largely similar. We also obtain similar results when we include a polynomial in ln tweets or use tweets in levels.

²⁶There might be heterogeneity in the duration of article popularity among readers based on its partisanship. For example, NYT readers may have a more sustained interest in pro-Democrat articles. In this case, more liberal articles on the NYT would see a slower decay in tweet counts, and similarly for conservative articles on the WSJ. Editors may therefore keep more partisan articles on the homepage longer to satisfy anticipated reader demand rather than due to owner- or firm-level ideological motives. However, Appendix D demonstrates that articles, regardless of their liberal or conservative tone, exhibit a consistent demand decay trajectory.

slant-duration correlations are stronger (Appendix F).

One coefficient that appears anomalous is the NYT Feature estimate on lagged tweets, where the hazard ratio exceeds one (1.121), implying faster removal when recent tweets are higher.²⁷ A plausible interpretation is operational: for softer Feature content, high early engagement may induce editors to rotate follow-up stories or allocate premium homepage slots to new items once interest is revealed.

More generally, interpretation should allow for mechanisms beyond a firm-level ideological strategy. Patterns can also arise from decentralized editor-level choices, heterogeneous forecasts, or systematic forecast errors at the desk level. For example, some story types may be repeatedly over- or under-predicted. These mechanisms can produce persistent slant-duration correlations without requiring explicit owner directives.

While not central to our analysis, the remaining covariates shed light on editorial decision-making. The sentiment of an article, measured by abstract sentiment scores, is strongly linked to its duration on the homepage. For the NYT, a one standard deviation increase in sentiment scores results in a 2% decrease in the hazard rate, meaning more positively toned articles stay longer. This effect is even more pronounced for the WSJ, where a similar increase corresponds to a 4% decrease in the hazard rate. Recall that more negative articles get placed on the homepage, so this means the longer duration ones are more representative.

Additionally, the publication of new articles within a given hour increases the hazard rate, suggesting that new content accelerates the removal of existing articles from the homepage. This is not a mechanical relationship: unlike a print newspaper, there is no fixed space constraint on the digital homepage, allowing NYT and WSJ to vary the number of articles over time.

All of these results extend to the positioning of articles *within* the homepage (Appendix G has details). The topmost articles are the first ones a reader encounters and so are the

²⁷See NYT Feature column in Table 6. Hazard ratios above one imply shorter, not longer, expected homepage survival.

most prominent. We can extract an article’s homepage position using its API ordering or URL suffix. The same qualitative patterns continue to hold when we restrict attention to article duration on the top section of the homepage. At the keyword level, partisan correlations remain visible in both papers even when controlling for tweet count, though these patterns are best viewed as evidence on editorial prominence conditional on observed demand rather than as stand-alone proof of supply-side bias.

A further concern is that the mix of stories can differ across the two papers. To address this, we also compare matched stories that cover the same event on the same day. The matched-story exercise yields qualitatively similar patterns: differences in relative homepage duration still co-move with relative tweet performance and with article slant. Because the exercise is mainly a robustness and trimming device rather than a central identification strategy, we move the full design, table, and section breakdown to Appendix H. The appendix results are informative, but we do not view them as eliminating all alternative mechanisms.

The appendix also probes the main results along several dimensions. Appendix B validates the article slant score using the ideological composition of retweeters. Appendixes D–F test whether demand decay varies with slant, replace total tweets with positive tweets, allow a polynomial in lagged log tweets, and estimate the model across sections and political tags. Appendix G then restricts attention to the most prominent homepage positions. Across these exercises, the NYT slant-duration association generally remains, while the pooled WSJ relationship is weaker and appears mainly in selected sections and tags. The demand-decay estimates show no systematic association between slant and tweet trajectories. Finally, Appendix J compares homepage editorial choices with print-edition and podcast decisions. Print selection exhibits slant correlations in several NYT sections, whereas podcast selection shows weaker slant relationships. These cross-format differences are consistent with supply-driven models, but we treat them as suggestive rather than decisive because audiences, editorial teams, and production constraints differ across formats.

6. Conclusion

This study contributes a new way to measure editorial emphasis: article-level homepage duration. The approach is portable across outlets, relies on commonly available inputs, and complements existing slant measures by focusing on prominence rather than only textual orientation.

Using over 100,000 NYT and WSJ articles, we document robust demand effects and meaningful slant-duration associations, especially for NYT. Additional matched-story comparisons and supplementary cross-platform evidence in the appendix point in similar directions, with print selection showing stronger slant correlation than podcast selection. Taken together, these facts are difficult to reconcile with a purely demand-based interpretation, but they do not by themselves isolate a single mechanism.

At the same time, our evidence is not definitive causal proof that supply dominates demand. A profit-maximizing niche-catering mechanism, audience-composition differences across platforms, and decentralized editor-level forecasting errors can also generate part of the observed patterns. We therefore view the paper’s main value as methodological and descriptive: it provides a transparent measurement framework and benchmark magnitudes that future work can use to isolate mechanisms more cleanly.

This study has limitations. We focus on two national newspapers, and our interpretation still depends on assumptions about demand proxies and comparable content environments. Future work can extend the design to other outlets, periods, and contexts (including historical archives), and can bring richer audience-level data to test competing mechanisms directly.

Appendix A Word Clouds of Other News Sections

There is a consistent set of words in articles categorized as either liberal or conservative across the newspapers. This gives some confidence in the machine learning algorithm used to generate the scores.

An examination of word clouds for news articles within the “World” section in Figure A1 reveals distinct patterns related to geopolitical events. Notably, the Ukraine conflict is frequently mentioned in both newspapers. While this theme appears in both liberal and conservative articles, it is more prominent in liberal articles in both newspapers, suggesting a broader focus or specific narrative angle in liberal coverage of the Ukraine situation.

Figure A1: Word Clouds of World News Articles' Abstracts



In relation to China, keywords such as “China”, “Chinese”, “Beijing”, “Xi”, and “Communist” appear more frequently in conservative articles across both newspapers, reflecting the hawkish stance against China among Republican politicians.

Analyzing word clouds for “Business” section articles in Figure A2 reveals similar

Figure A2: Word Clouds of Business News Articles' Abstracts



thematic trends. The term “China” is a recurrent motif in conservative articles, consistent with observations from the “World” section. This suggests a sustained conservative focus on China-related issues across both global and business contexts.

Additionally, the term “inflation” is prominent in conservative articles in both newspapers. This reflects Republican concerns about the negative effects of rising interest rates by the Federal Reserve. The focus on inflation in these conservative articles highlights partisan concerns about economic impacts, underscoring the interplay between fiscal policy perspectives and media representation.

Appendix B Validation of Article Slant Using Retweeter Ideology

To address concerns about the “black box” nature of text-based partisan measures, we provide an alternative validation using the ideological composition of Twitter users who retweet each article. If our article-level slant score captures ideological orientation, then more left-leaning articles should be disproportionately retweeted by users with more left-leaning ideological profiles. In short, this is a revealed preference measure of each article’s partisan slant.

Our revealed preference measure is based on a media bias measure, [Blindspotter](#).²⁸ Blindspotter identifies news stories that receive lopsided coverage from either left-leaning or right-leaning media outlets, with the goal of helping readers identify “blind spots” (stories covered by only one side of the political spectrum) in their news consumption. Their measure is based on which members of a large panel with known political views interact with the story. It starts by creating a political bias distribution (percent L/C/R) for millions of Twitter accounts based on the type of news articles each one engages with.²⁹ These biases are revised as the accounts engage with additional articles.

Through a special agreement with Ground News, the company that created Blindspotter, we obtained access to their full database of Twitter accounts over the period of our analysis.³⁰ We then matched these accounts to our database, which included all tweets involving a NYT and WSJ article over our sample period. For the matched accounts, we generated a complete list of retweets of articles from either source. Retweets were treated as a positive endorsement of the article’s content. If an account was critical of the article, the user could add text indicating the issue rather than retweet. Our bias measure was based on the distribution of accounts that retweeted the article: the share of retweeting

²⁸Full details on Blindspotter methodology are available at <https://ground.news/blindspotter/methodology>.

²⁹It uses a ranking of political slant based on the average of three widely used sources: <https://mediabiasfactcheck.com>, <https://allsides.com>, and <https://adfontesmedia.com>.

³⁰The database has 3.4 million accounts. These are accounts with enough usage to create a distribution, with each account having an average of 70 news engagements.

users whose BlindSpotter profiles indicated left-leaning news consumption. The results continued to hold when we used the article-level ideology average of all retweeting users. We also considered weighted versions of this measure, weighting either by the number of news interactions observed in BlindSpotter or by the follower counts of the retweeting users, so that more active news consumers or more influential users received greater weight.

There are two key assumptions in the approach. First, the ideology score for each Twitter account is accurate. Since Ground News has access to each account’s full Twitter history, this essentially boils down to whether the underlying ideology rating sites (in footnote 29) are accurate. These sites are widely used in media studies, so this seems a reasonable assumption. Second, retweets measure endorsement of a topic. Since no additional text or context is added, retweets are designed to amplify an initial post (if a user wanted to criticize or provide context they could instead quote tweet). As NPR’s Standards and Practices editor Mark Memmott wrote in an internal staff memo in 2014: “retweets should be viewed AS endorsements”.³¹

We then validate our machine learning slant score with article-level regressions of the form,

$$\text{article slant score}_i = \beta \times \% \text{left among retweeters}_i + u_i$$

If the two measures capture the same partisan slant, the relationship should be strong (R^2 near one and β approaching one). Because the BlindSpotter data end in April 2022, not all retweeting users can be matched to ideology estimates. We therefore restrict attention to articles for which at least 80% of retweets can be matched to BlindSpotter users, ensuring that the ideological composition measure is reliably estimated. Table B1 shows that the regressions have very high explanatory power across both newspapers and

³¹<https://www.npr.org/sections/publiceditor/2014/07/31/336921115/the-ethics-of-retweeting-and-whether-it-amounts-to-endorsement>

weighting schemes: the revealed preference measure explains between 70 and 90% of the variation in the machine learning approach we originally used. The regression coefficient is also between 0.8 and 1.0. These results indicate there is a roughly one-for-one relationship between the two measures.

Table B1: Validation of Slant Score Using Ideological Composition of Retweeters

	(1)	(2)	(3)	(4)	(5)	(6)
	NYT: % left	NYT: % left (BS wt.)	NYT: % left (Followers wt.)	WSJ: % left	WSJ: % left (BS wt.)	WSJ: % left (Followers wt.)
% left	0.844*** (0.002)	0.800*** (0.002)	0.828*** (0.002)	1.014*** (0.012)	1.006*** (0.013)	0.968*** (0.013)
Adjusted R-squared	0.880	0.878	0.868	0.740	0.716	0.702
N	30715	30544	30709	2518	2454	2516

Standard errors in parentheses

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Appendix C Impact of Homepage Presence on Article Popularity (Tweet Count)

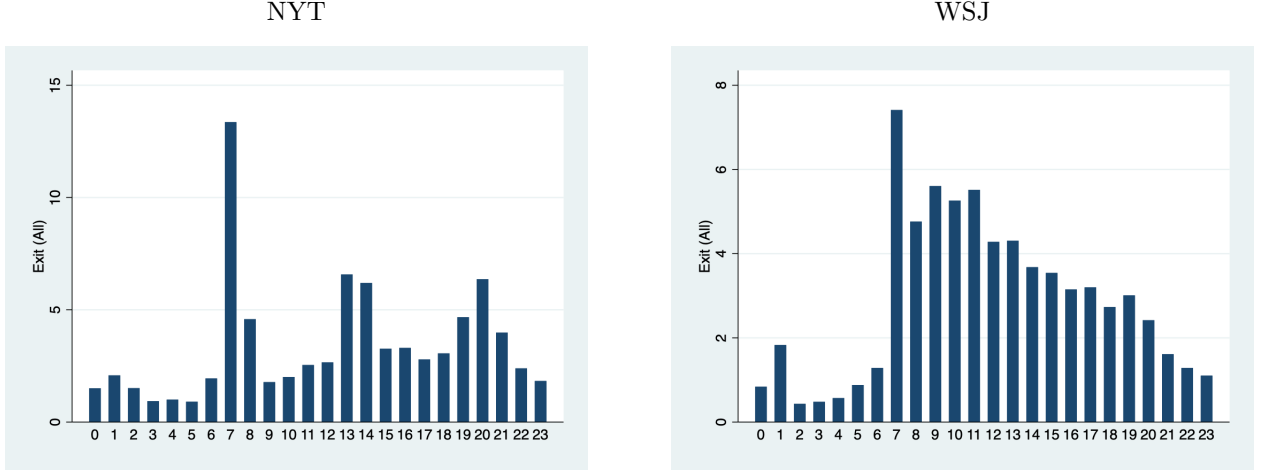
One way that the digital homepage matters is through increasing readership of articles.³² Establishing such a causal relationship, however, is challenging. A straightforward regression with tweet count (our measure of popularity) as the dependent variable and homepage presence as an independent variable is problematic due to endogeneity. Articles are not randomly placed on the homepage. They are selected based on various unobserved factors, potentially confounding the relationship between homepage presence and popularity.

We address this endogeneity issue using an instrumental variable (IV) approach. Our chosen IV leverages a pattern observed in the timing of article withdrawals from the homepage.

Figure C1 illustrates the average number of articles removed from the homepage by time of day. These patterns, shown for the NYT on the left and the WSJ on the right, highlight certain hours where pronounced homepage updates occur. The spikes are driven by routine editorial practices rather than article demand: they occur when there is a changeover in editors responsible for the homepage.

³²The homepage, often seen as the digital equivalent of a newspaper's front page, is important in shaping readers' perceptions and guiding their attention. Editorial decisions about which stories remain and which are withdrawn may reflect editorial priorities and influence news dissemination and reception.

Figure C1: Article Withdrawals by Hour of Day (Eastern Time)



For the NYT, notable spikes in article withdrawal occur at 7am, 8am, 1pm, 2pm, 7pm, and 8pm. For the WSJ, spikes are primarily at 7am and 1am. We use these specific hours as instrumental variables in our analysis to address endogeneity and accurately estimate the impact of homepage presence on article popularity, measured by tweet counts.

In order to examine the impact of homepage presence on the popularity of articles as measured by tweet counts, we employ the following regression specification:

$$\ln Y_{it} = \beta_0^H \text{Homepage}_{it} + \beta_1^H X_{it} + \phi_i + \varepsilon_{it}, \quad (\text{C1})$$

where $\ln Y_{it}$ represents the natural logarithm of the zero-inflated tweet counts pertaining to article i at time t .

Our primary independent variable of interest, Homepage_{it} , is a binary indicator capturing whether article i is prominently displayed on the homepage at time t . This serves as our key measure to understand the immediate influence of homepage placement on article popularity as gauged by tweet counts. Recognizing the potential endogeneity of Homepage_{it} , which could confound our estimates, we employ an instrumental variable approach. Specifically, we use $\text{WithdrawalHours}_{it}$ as an instrument, which is a binary indicator taking the value of one if time t coincides with the specific hours during which

the newspaper undertakes its routine editorial practice of refreshing its homepage. This instrumental strategy allows us to exploit variations induced by these editorial practices to identify the causal effect of homepage presence on tweet counts.

The vector X_{it} encompasses a set of controls that account for various determinants of article popularity. This includes the lagged logarithm of the zero-inflated tweet counts, denoted as $\ln Y_{it-1}$, to control for the article’s inherent virality in the preceding time period. Moreover, we adjust for the number of hours since the article’s publication to account for potential decay effects in attention and visibility over time. To further refine our estimations, we introduce a series of time fixed effects. These include controls for specific times of the day, days of the week, and months, capturing cyclical patterns in readership and online activity.

Lastly, to account for any persistent differences across articles that might be influencing their tweet counts, we incorporate article-specific fixed effects, ϕ_i . These effects mitigate biases arising from unobserved heterogeneity across articles that remain constant over time.

Our findings in Table C1 analyze the impact of homepage placement on tweet counts for articles from both newspapers. To maintain consistency with Section 5, our sample includes only articles that were on the homepage in the preceding hour ($t - 1$). This criterion aligns our sample with the main text focus on article homepage longevity.

Table C1: Ln Tweets Regression with Article F.E.

	NYT				WSJ		
	News	Opinion	Feature	All	News	Feature	All
Second Stage. Dependent variable is Ln Tweets							
Homepage	0.320*** (0.0386)	0.764*** (0.171)	0.263*** (0.0215)	0.349*** (0.0241)	1.496*** (0.144)	2.070*** (0.319)	1.622*** (0.131)
Ln Tweets (Lag)	0.407*** (0.00172)	0.413*** (0.00230)	0.270*** (0.00265)	0.400*** (0.00119)	0.292*** (0.00403)	0.213*** (0.00928)	0.278*** (0.00374)
Hours Published	-0.0268*** (0.000487)	-0.0178*** (0.000497)	-0.0162*** (0.000332)	-0.0207*** (0.000181)	-0.0112*** (0.00118)	-0.00392* (0.00203)	-0.00916*** (0.000996)
Observations	384512	223509	137959	745980	141749	46921	188670
Time-of-Day FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Day FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
First Stage. Dependent variable is Homepage							
Withdrawal Hours	-0.0716*** (0.00109)	-0.0190*** (0.000807)	-0.170*** (0.00179)	-0.0770*** (0.000706)	-0.0497*** (0.00251)	-0.0432*** (0.00468)	-0.0482*** (0.00221)
Ln Tweets (Lag)	0.0235*** (0.000563)	0.00680*** (0.000474)	0.0230*** (0.00127)	0.0222*** (0.000394)	0.0189*** (0.000987)	0.0212*** (0.00204)	0.0199*** (0.000890)
Hours Published	-0.0116*** (0.0000743)	-0.00283*** (0.0000331)	-0.00983*** (0.000123)	-0.00629*** (0.0000382)	-0.00787*** (0.0000994)	-0.00616*** (0.000163)	-0.00734*** (0.0000846)
Time-of-Day FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Day FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Standard errors in parentheses							
* $p < .1$, ** $p < .05$, *** $p < .01$							

The table is divided into two panels: the left for NYT results and the right for WSJ. Each panel is further split into two parts: the upper half presents the second-stage results, while the lower half outlines the first-stage findings.

In the first-stage results, both NYT and WSJ show a higher likelihood of article withdrawal during specific hours, consistent with our expectations based on routine editorial practices.

The second-stage results highlight the significant impact of homepage presence on an article's tweet counts. The magnitude varies notably between the two newspapers. For the NYT, homepage presence increases tweet counts by 35% on average, with op-ed articles seeing a 76% rise and feature articles a 26% increase. WSJ articles experience a much larger boost from homepage placement, with an average increase of 162%. News articles in WSJ see a 150% rise in tweet counts, while feature articles spike by over 200%.³³

³³A possible explanation for the smaller impact on NYT articles is the newspaper's effective use of

These estimates show that homepage presence has a robust and economically meaningful impact on an article's popularity as measured by Twitter activity.

other promotional channels, such as Facebook and Twitter, which may mitigate the relative benefits of homepage placement.

Appendix D Does Demand Decay Vary by Political Slant?

In our analysis of homepage duration in Table 6, there is a potential challenge to our interpretation. One might argue that the effects attributed to supply-side factors, measured by an article’s pro-Democrat score, could actually be driven by latent demand dynamics. For instance, articles with a pronounced liberal slant (in the case of NYT) or conservative slant (in the case of WSJ) might inherently attract greater sustained interest, leading to a slower decay in demand, even when prior tweet counts are equivalent. In this case our partisan variable need not reflect supply-side preferences but rather editor forecasts of future demand.

To address this concern, we investigate the temporal decay in tweet shares for articles with the following regression:

$$\ln Y_{it} - \ln Y_{it-1} = \beta Z_{it} + \epsilon_{it}, \quad (\text{D1})$$

where Y_{it} is the zero-inflated tweet count for article i at time t , and $\ln Y_{it} - \ln Y_{it-1}$ is the log difference in tweet counts, namely the decay in tweets. Z_{it} is a set of (potentially) time-varying controls for article i , including the number of hours since publication, the sentiment and pro-Democrat scores of the article, presence on the homepage, and number of new articles published by the newspaper at time t . Additionally, the model incorporates several fixed-effect dummies for the time of the article’s publication (hour of the day, day of the week, and year-month).

The estimated constants in Table D1 indicate that, regardless of the newspaper, tweets for most sections’ articles decline by 5.2% to 7.6% per hour in the initial stages after publication, with NYT op-ed articles as the only exception. Furthermore, as articles age, the rate of decay slows, as shown by the estimates on hours since publication.³⁴

³⁴Additionally, sentiment scores are not significantly associated with tweet decay. The only exception is NYT news articles, where those with a more positive tone experience a slower decay in tweets. Homepage presence and number of new articles published during that period have mixed effects.

Table D1: Log Difference (Tweets) Regressions

	NYT				WSJ		
	News	Opinion	Feature	All	News	Feature	All
Sentiment Scores (Abstract)	0.00224** (0.00102)	-0.000605 (0.00291)	-0.000386 (0.00122)	0.00125 (0.000763)	-0.000669 (0.00204)	0.00204 (0.00268)	0.0000995 (0.00163)
Pro-Dem. Scores (Abstract)	-0.000801 (0.00205)	0.00195 (0.00540)	0.000572 (0.00252)	-0.000214 (0.00154)	0.00227 (0.00406)	0.00530 (0.00541)	0.00319 (0.00327)
Homepage	0.0197*** (0.00134)	-0.0200*** (0.00300)	0.0227*** (0.00158)	0.0205*** (0.000952)	-0.0137*** (0.00231)	-0.0139*** (0.00299)	-0.0141*** (0.00183)
Hours Published	0.000850*** (0.0000364)	-0.00152*** (0.000101)	0.00181*** (0.0000364)	0.00105*** (0.0000253)	0.00122*** (0.0000757)	0.000911*** (0.0000882)	0.00112*** (0.0000580)
# of New Articles	0.00107*** (0.000163)	0.0133*** (0.00105)	0.00195*** (0.000172)	0.000933*** (0.000118)	-0.000130 (0.000415)	-0.00116** (0.000569)	-0.000825** (0.000369)
Constant	-0.0670*** (0.00368)	0.00134 (0.0102)	-0.0755*** (0.00410)	-0.0689*** (0.00296)	-0.0742*** (0.00654)	-0.0516*** (0.00808)	-0.0711*** (0.00516)
Observations	2214816	324240	1305408	3844464	497136	236352	733488
Hour FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Day FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Standard errors in parentheses

* $p < .1$, ** $p < .05$, *** $p < .01$

The most important result is the test of the demand expectations hypothesis. There is little support for this in the estimates: pro-Democrat scores show no statistically significant association with the rate of tweet decay. In other words, articles with a liberal slant in NYT or a conservative slant in WSJ do not exhibit a distinct deceleration in decay compared to other articles. Article partisanship does not predict tweet trajectories. This makes a pure latent-demand-decay explanation less compelling, though it does not rule out other mechanisms such as editorial forecasting, audience sorting, or format-specific decision rules.

Appendix E Survival Analysis with Positive Tweets

The primary results from our survival analysis show that slant-duration correlations remain after accounting for tweet counts, our proxy for demand-side engagement. However, not all tweets may reflect genuine demand for the articles. For example, tweets criticizing the reporting might not correlate with increased views or positive engagement.

To address this concern, we conducted a robustness check by considering only positive tweets instead of the total tweet count (we use our machine learning-based sentiment score to classify tweets as positive or not, see footnote 18). The results of this reanalysis, detailed in Table E1, remain consistent with our initial findings. The hazard rate appears even more sensitive to the number of positive tweets than the overall tweet count. The broader interpretation is unchanged: more liberal articles continue to stay longer on the NYT homepage after adjusting for demand-side factors, while the WSJ shows no significant pooled association between slant and homepage duration.

Table E1: Survival Analysis of Homepage (Hazard Ratios)

	NYT				WSJ		
	News	Opinion	Feature	All	News	Feature	All
Ln Tweets	0.903*** (0.00809)	0.540*** (0.0227)	1.250*** (0.0267)	0.925*** (0.00752)	0.734*** (0.0226)	0.604*** (0.0395)	0.702*** (0.0193)
Sentiment Scores (Abstract)	0.958*** (0.0112)	0.967 (0.0320)	0.984 (0.0202)	0.961*** (0.00931)	0.886*** (0.0204)	0.999 (0.0394)	0.921*** (0.0178)
Pro-Dem. Scores (Abstract)	0.916*** (0.0227)	0.878** (0.0531)	0.876*** (0.0390)	0.900*** (0.0183)	0.987 (0.0472)	0.950 (0.0800)	0.955 (0.0388)
# of New Articles	1.042*** (0.00198)	1.257*** (0.00917)	1.015*** (0.00339)	1.023*** (0.00103)	1.073*** (0.00436)	1.044*** (0.00779)	1.041*** (0.00208)
Observations	384488	223499	137890	745877	141491	46784	198575
Hour FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Day FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes

p-stats are for hazard ratio differences from one.

Estimates for α , λ from (1) are omitted.

Exponentiated coefficients; Standard errors in parentheses

* $p < .1$, ** $p < .05$, *** $p < .01$

We also re-estimate the survival model using a more flexible control for demand by including a polynomial in lagged log tweets. Table E2 shows that the main pattern is largely unchanged. The NYT pro-Democrat coefficients remain below one and close to the baseline estimates, while the pooled WSJ coefficient remains close to one and

statistically indistinguishable from zero. Allowing for this polynomial refinement of the demand control therefore does not materially alter the substantive conclusions.

Table E2: Survival Analysis of Homepage with Polynomial Tweets Controls (Hazard Ratios)

	NYT				WSJ		
	News	Opinion	Feature	All	News	Feature	All
Ln Tweets	1.093*** (0.0124)	0.668*** (0.0230)	1.119*** (0.0224)	1.064*** (0.00980)	1.216*** (0.0316)	1.002 (0.0506)	1.123*** (0.0247)
Ln Tweets Sq	0.972*** (0.00246)	1.035*** (0.0113)	1.001 (0.00590)	0.980*** (0.00214)	0.920*** (0.00779)	0.928*** (0.0189)	0.932*** (0.00695)
Sentiment Scores (Abstract)	0.952*** (0.0112)	0.934** (0.0309)	0.997 (0.0205)	0.957*** (0.00929)	0.871*** (0.0200)	0.972 (0.0382)	0.906*** (0.0175)
Pro-Dem. Scores (Abstract)	0.914*** (0.0226)	0.892* (0.0540)	0.874*** (0.0388)	0.899*** (0.0183)	0.976 (0.0466)	0.932 (0.0783)	0.948 (0.0385)
# of New Articles	1.042*** (0.00198)	1.254*** (0.00912)	1.015*** (0.00339)	1.023*** (0.00103)	1.063*** (0.00343)	1.087*** (0.00828)	1.043*** (0.00203)
Observations	384488	223499	137890	745877	142026	47038	199977
Hour FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Day FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes

p-stats are for hazard ratio differences from one.

Estimates for α , λ from (1) are omitted.

Exponentiated coefficients; Standard errors in parentheses

* $p < .1$, ** $p < .05$, *** $p < .01$

Appendix F Survival Analysis by Sections and News Tags

To provide deeper insights into how demand proxies and article slant relate to homepage duration, we divide our samples by major news sections and tags (keywords).

Table F1 presents the results by major news sections, revealing several interesting patterns.

First, higher tweet counts, which represent demand-side factors, typically reduce exit from the homepage for both newspapers. This effect is stronger in the news sections compared to the feature sections.

Second, political slant is correlated with homepage survival in several NYT sections, consistent with the pooled results in Table 6. Liberal-leaning articles tend to remain longer on the NYT homepage across several sections, with the notable exception of the “Style” section. This effect is particularly pronounced in the “Arts” section. In contrast, for the WSJ, an article’s political slant generally does not have a statistically significant association with its homepage survival duration. One exception is the “US” section, where articles with a pro-Democrat slant tend to exit the homepage.³⁵

We can also focus on sections that are relatively unique to each newspaper. In the WSJ’s “C Suite” section, liberal-leaning articles are far more likely to exit the homepage in any period (though the parameter is not statistically significant). One interpretation is that specialized content areas permit more editorial discretion, but the estimates are imprecise and can also reflect section-specific audiences or workflows. Conversely, in the NYT’s “NY Region” section, evidence for pronounced slant-duration correlations is weaker, although the limited sample size for this section might affect the findings.

³⁵Liberal-leaning articles in the “World”, “Arts”, and “Style” sections tend to stay longer on the homepage. However, the parameters are not statistically significant.

Table F1: Survival Analysis of Homepage by Sections (Hazard Ratios)

	NYT						
	Business	US	World	NY Region	Arts	Life	Style
Ln Tweets	0.928*** (0.0125)	0.952*** (0.00621)	0.963*** (0.00894)	0.958 (0.0557)	1.183*** (0.0186)	1.099*** (0.0191)	1.034 (0.0307)
Sentiment Scores (Abstract)	0.948 (0.0349)	1.063*** (0.0198)	0.877*** (0.0209)	0.971 (0.141)	1.031 (0.0336)	1.038 (0.0381)	0.918 (0.0569)
Pro-Dem. Scores (Abstract)	0.979 (0.0682)	0.889*** (0.0337)	0.751*** (0.0418)	1.040 (0.321)	0.654*** (0.0471)	0.838** (0.0683)	1.416*** (0.171)
# of New Articles	1.051*** (0.00559)	1.042*** (0.00311)	1.048*** (0.00417)	1.029 (0.0213)	1.020*** (0.00543)	0.999 (0.00661)	1.018** (0.00882)
Observations	43975	170752	82492	3862	52627	43795	18406

	WSJ						
	Business	US	World	C Suite	Arts	Life	Style
Ln Tweets	0.842*** (0.0195)	0.850*** (0.0171)	0.814*** (0.0156)	1.038 (0.191)	0.937 (0.111)	0.943 (0.0339)	1.210*** (0.0820)
Sentiment Scores (Abstract)	0.989 (0.0518)	0.997 (0.0435)	1.012 (0.0436)	0.781 (0.178)	0.942 (0.102)	0.930 (0.0555)	0.974 (0.127)
Pro-Dem. Scores (Abstract)	1.083 (0.102)	1.282*** (0.118)	0.869 (0.0864)	2.103 (0.981)	0.732 (0.197)	1.004 (0.133)	0.843 (0.223)
# of New Articles	1.066*** (0.00882)	1.085*** (0.00883)	1.074*** (0.00874)	1.073** (0.0370)	1.031 (0.0222)	1.061*** (0.0119)	1.060*** (0.0225)
Observations	33652	33641	32087	23853	4000	20204	5533
Hour FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Day FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes

p-stats are for hazard ratio differences from one.

Estimates for α , λ from (1) are omitted.

Exponentiated coefficients; Standard errors in parentheses

* $p < .1$, ** $p < .05$, *** $p < .01$

The role of article sentiment also varies between the two newspapers (though the parameters are not significant). For the WSJ, sentiment has little impact on article duration on the homepage, except in the “C Suite” section where more positive articles tend to stay longer. In contrast, sentiment plays a more significant role in the NYT, particularly in the news section. Generally, articles with a more positive tone have longer survival times on the homepage, except in the U.S. news section where more negative news tends to stay longer.

Table F2 presents homepage survival estimates for subsets of articles the newspapers tag with keywords corresponding to five major political topics. The topics are: “Politics”, including news about political parties and elections; “Coronavirus”, covering the

pandemic and vaccinations; “Race/Gender”, addressing issues such as discrimination and significant events like the George Floyd incident; “Abortion”, involving news related to abortion, including Supreme Court decisions; and “Immigration”, encompassing news on immigration issues.

Table F2: Survival Analysis of Homepage by Major Tags (Hazard Ratios)

	NYT					WSJ				
	Politics	Coronavirus	RaceGender	Abortion	Immigration	Politics	Coronavirus	RaceGender	Abortion	Immigration
Ln Tweets	0.937*** (0.00665)	1.079*** (0.0135)	1.102*** (0.0145)	0.981 (0.0288)	0.975 (0.0248)	0.971*** (0.00967)	0.878*** (0.0327)	0.905*** (0.0305)	0.749** (0.0894)	0.948 (0.142)
Sentiment Scores (Abstract)	1.012 (0.0210)	1.029 (0.0381)	0.724*** (0.0257)	1.106 (0.104)	0.865** (0.0584)	0.963 (0.0220)	0.889 (0.0974)	0.725*** (0.0582)	1.195 (0.334)	0.856 (0.275)
Pro-Dem. Scores (Abstract)	0.862*** (0.0347)	0.887* (0.0627)	1.107 (0.0877)	0.621** (0.123)	0.577*** (0.0871)	1.216*** (0.0573)	0.997 (0.189)	1.749*** (0.265)	1.139 (0.494)	1.061 (0.695)
# of New Articles	1.023*** (0.00240)	1.023*** (0.00372)	1.017*** (0.00424)	1.015 (0.0102)	1.015* (0.00791)	1.038*** (0.00244)	1.044*** (0.0106)	1.048*** (0.00830)	1.070*** (0.0247)	1.039 (0.0300)
Observations	133830	42765	43969	9245	12121	157417	7888	12823	2299	1170
Hour FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Day FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

p-stats are for hazard ratio differences from one.

Estimates for α , λ from (1) are omitted.

Exponentiated coefficients; Standard errors in parentheses

* $p < .1$, ** $p < .05$, *** $p < .01$

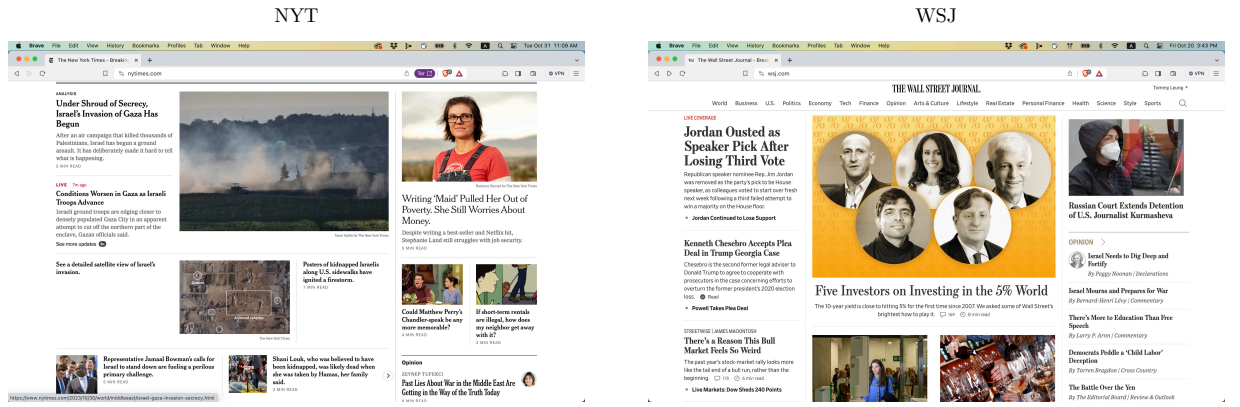
Consistent with our prior findings, audience engagement, as indicated by tweet counts, is a key factor for longer homepage duration across most topic tags and for both newspapers. The exceptions are articles tagged with “Coronavirus” and “Race/Gender” in the NYT.

Political slant is also correlated with article homepage survival within several topic tags. The “Politics” category is representative: more liberal articles in the NYT and more conservative articles in the WSJ tend to stay longer on the homepage. A one-standard-deviation increase in the pro-Democrat score results in a 2.4% decrease in the hazard rate for NYT articles and a 4% increase for WSJ articles. Similar patterns are observed in other topic tags, particularly “Abortion”. As elsewhere in the paper, these tag-level estimates are best viewed as conditional relationships and may reflect multiple mechanisms beyond owner-level ideological supply preferences.

Appendix G Survival Analysis on Leading News

While our prior analysis focused broadly on article dynamics across the entire homepages of major newspapers, this section narrows the focus to leading news, the articles occupying the most prominent slots at the top of the homepage. These articles often set the day’s narrative and attract the most immediate attention from visitors.³⁶ Figure G1 illustrates the positions of leading news in the NYT and WSJ, respectively.

Figure G1: Leading News on NYT and WSJ Homepage



To identify the “leading news” in the WSJ, the URL structure of WSJ articles provided a convenient marker for discerning an article’s position on the homepage. Specifically, appended information at the end of each URL demarcates its location. For instance, URLs tagged with “hp_lead” correspond to articles placed among the top 10–11 slots on the homepage, effectively the marquee stories of the hour. Another category, “hp_featst”, denotes articles positioned right below the primary “hp_lead” cluster.³⁷ For our analysis, we treated articles marked as “hp_lead” as the leading news for that particular hour in the WSJ.

³⁶These articles get more attention than others on the homepage. For example, 5.1% of the NYT’s ten most viewed articles each hour are from the top of the homepage, while the far higher number of articles on the remainder of the homepage make up 3.0% (the values are 7.2% / 2.8% for top ten shares and 2.8% / 2.3% for top ten emails).

³⁷The other major position tags in the URLs include “hp_jr”, “hp_lista”, “hp_listb”, “hp_listc”, “hp_minor”, “hp_lead_feature_below_a”, and “wsj_hp_buyside”.

Identifying leading news for the NYT required a different approach due to the lack of position-specific markers in their URLs. We used the order in which articles were scraped from the API, where the first ten URLs in the sequence were typically those with the most prominent positions on the NYT homepage. These were classified as leading news for the NYT.³⁸

It is important to note that the bulk of these leading articles are from the News category. As such, our examination in this section excludes Opinion and Feature articles, concentrating solely on the News pieces to ensure consistency and specificity in the findings. Table G1 presents separate survival (at the leading news position of the homepages) estimates for each news section.

Table G1: Survival Analysis of Leading News by Major News Sections

	NYT			WSJ		
	Business	US	World	Business	US	World
Ln Tweets	0.875*** (0.0165)	0.888*** (0.00797)	0.875*** (0.0122)	0.819*** (0.0246)	0.843*** (0.0226)	0.872*** (0.0244)
Sentiment Scores (Abstract)	1.118** (0.0602)	1.158*** (0.0307)	0.987 (0.0356)	0.934 (0.0672)	1.086 (0.0745)	1.074 (0.0757)
Pro-Dem. Scores (Abstract)	1.009 (0.0987)	0.856*** (0.0468)	0.650*** (0.0550)	0.980 (0.132)	0.950 (0.137)	1.610*** (0.268)
# of New Articles	1.054*** (0.00808)	1.031*** (0.00432)	1.032*** (0.00609)	1.096*** (0.0123)	1.074*** (0.0127)	1.058*** (0.0128)
Observations	17901	70081	35486	13909	12347	12037
Hour FE	Yes	Yes	Yes	Yes	Yes	Yes
Day FE	Yes	Yes	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes	Yes	Yes

p-stats are for hazard ratio differences from one.

Estimates for α , λ from (1) are omitted.

Exponentiated coefficients; Standard errors in parentheses

* $p < .1$, ** $p < .05$, *** $p < .01$

Consistent with our previous findings, political slant significantly influences the editorial decisions for leading news selection, even after accounting for demand-side dynamics. In the NYT, liberal-leaning U.S. and World news articles tend to remain in top homepage positions longer. A one-standard-deviation increase in pro-Democrat scores decreases the hazard rates by 3% for U.S. news and 7% for World news, indicating a lower likelihood

³⁸The analysis excludes Opinion, which has a fixed position below leading news in the NYT and was not scraped for the WSJ.

of swift replacement.

Conversely, the WSJ shows a distinct trend in its World news section, where conservative-leaning articles are more resilient in top positions. A one-standard-deviation rise in pro-Democrat scores corresponds to a 12% increase in hazard rates, suggesting that liberal-leaning World news articles in the WSJ are more likely to be quickly replaced.

Table G2 splits the samples by major news tags (these keywords are described in more detail in Appendix F). The results show that on the NYT homepage, articles tagged with “Politics”, “Coronavirus”, “Abortion”, or “Immigration” that have a liberal slant tend to dominate the top spots for extended periods. Conversely, in the WSJ, leading news articles tagged with “Politics” or “Race/Gender” that have a conservative orientation have a longer duration on their prominent homepage positions.

Table G2: Survival Analysis of Leading News by Major Tags (Hazard Ratios)

	NYT					WSJ				
	Politics	Coronavirus	Race/Gender	Abortion	Immigration	Politics	Coronavirus	Race/Gender	Abortion	Immigration
Ln Tweets	0.937*** (0.00665)	1.079*** (0.0135)	1.102*** (0.0145)	0.981 (0.0288)	0.975 (0.0248)	0.971*** (0.00967)	0.878*** (0.0327)	0.905*** (0.0305)	0.749** (0.0894)	0.948 (0.142)
Sentiment Scores (Abstract)	1.012 (0.0210)	1.029 (0.0381)	0.724*** (0.0257)	1.106 (0.104)	0.865** (0.0584)	0.963 (0.0220)	0.889 (0.0974)	0.725*** (0.0582)	1.195 (0.334)	0.856 (0.275)
Pro-Dem. Scores (Abstract)	0.862*** (0.0347)	0.887* (0.0627)	1.107 (0.0877)	0.621** (0.123)	0.577*** (0.0871)	1.216*** (0.0573)	0.997 (0.189)	1.749*** (0.265)	1.139 (0.494)	1.061 (0.695)
# of New Articles	1.023*** (0.00240)	1.023*** (0.00372)	1.017*** (0.00424)	1.015 (0.0102)	1.015* (0.00791)	1.038*** (0.00244)	1.044*** (0.0106)	1.048*** (0.00830)	1.070*** (0.0247)	1.039 (0.0300)
Observations	133830	42765	43969	9245	12121	157417	7888	12823	2299	1170
Hour FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Day FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

p-stats are for hazard ratio differences from one.

Estimates for α , λ from (1) are omitted.

Exponentiated coefficients; Standard errors in parentheses

* $p < .1$, ** $p < .05$, *** $p < .01$

Appendix H Matched Stories Across NYT and WSJ

One concern with the survival analysis in the main text is that the mix of stories can differ between the papers. This appendix therefore compares homepage duration for matched articles covering the same story on the same day. We view this exercise as a useful robustness check because it narrows differences in topic mix and timing, but it does not eliminate all alternative mechanisms.

To ensure a precise comparison, we identified pairs of articles from each newspaper that reported the same news on the same day by calculating the cosine similarity between their headlines and abstracts (articles can be part of multiple news topic clusters and so matched to more than one article from the other paper).³⁹ Articles were considered to cover the same news if their cosine similarity score exceeded 0.2.⁴⁰

For the period October 2022 to May 2023, when data are available for both papers, we identified 2,031 pairs of news articles that appeared on both newspapers' homepages. This set includes 1,817 unique NYT articles (10.4% of 17,452 sampled) and 1,706 unique WSJ articles (11.9% of 14,379 sampled). In our matched sample, over 90% of the NYT articles and 80% of the WSJ articles are categorized as News articles, rather than Feature or Op-Ed articles. Figure H1 shows that within this News category, the predominant sections are Business, US, and World, collectively encompassing more than 70% of the identified articles.

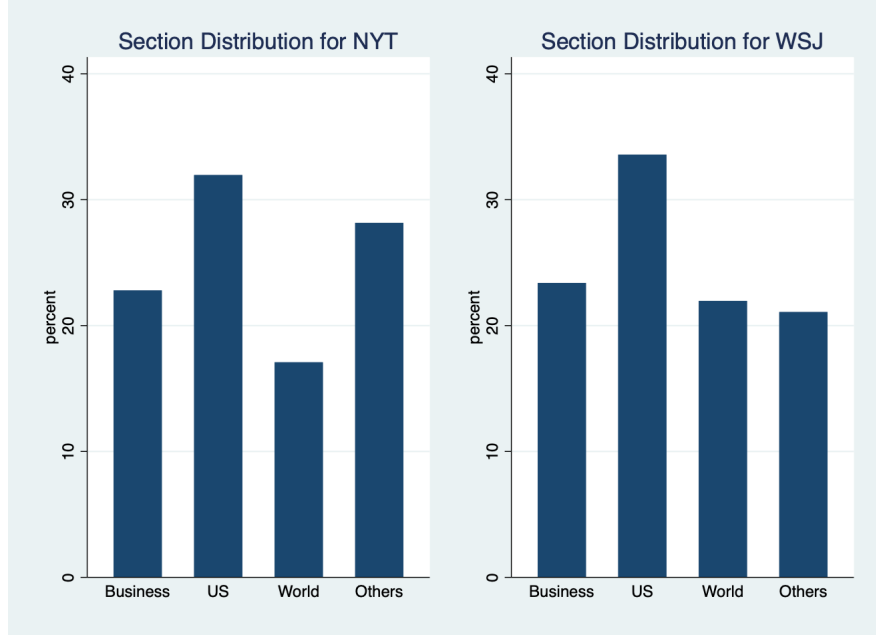
We used a simple ordinary least squares (OLS) regression to analyze the difference in survival length (hours) on the homepages of the NYT and WSJ,

³⁹Text preprocessing included lemmatization to group similar words and removal of stop words for enhanced matching accuracy.

⁴⁰Similar results were obtained with thresholds of 0.25 and 0.3, confirming the robustness of our matching criterion.

To more formally validate our approach, we had a research assistant manually match similar articles from the two newspapers on three dates. The manually matched sample closely resembled our sample obtained through textual analysis. In fact, our sample had fewer false positives.

Figure H1: Distribution of Sections Among Matched News Articles



$$\Delta_i = \delta X_i + u_i \quad (\text{H1})$$

where $\Delta_i \equiv \tau_i^{NYT} - \tau_i^{WSJ}$ is the difference in homepage duration for matched article i on the NYT relative to the WSJ. The X_i are covariates from the two articles such as the pro-Democrat scores of the articles, the difference in tweet counts within the first six hours of publication, and other time-fixed effects. Since pro-Democrat scores often differ between the two newspapers even for the same news, we consider different specifications that either average pro-Democrat scores from the articles in the two papers or use specific scores from either the NYT or WSJ. We also have specifications that restrict our sample to article pairs with close pro-Democrat scores.

The results in Table H1 reveal two primary findings. First, tweet count differences significantly influence survival duration; articles with more NYT- than WSJ-related tweets remain longer on the NYT homepage. Second, even after adjusting for tweet count differentials, more liberal articles consistently show extended survival on the NYT homepage compared to the WSJ. For example, a one-standard-deviation increase in average

pro-Democrat scores (column one) is associated with an additional hour on the NYT homepage relative to the WSJ. This pattern holds across various specifications, including average pro-Democrat scores (column one), individual scores from each newspaper (columns 2–3), and stronger results (in terms of economic magnitude) among articles with closely matched pro-Democrat scores (columns 4–5). These findings are consistent with the main-text patterns, though they should still be interpreted cautiously because even matched stories can face different editors, audience expectations, and packaging decisions across outlets.

Table H1: OLS Regression of Survival Difference (NYT - WSJ)

	(1) Avg Pro-Dem Score	(2) NYT Pro-Dem Score	(3) WSJ Pro-Dem Score	(4) Pro-Dem Dif. < 0.1	(5) Pro-Dem Dif. < 0.05
Pro-Dem. Scores (Abstract)	4.791*** (1.653)	2.424* (1.334)	3.958*** (1.360)	4.918* (2.574)	6.830* (3.940)
Tweets Dif. (NYT - WSJ)	0.000886*** (0.000250)	0.000904*** (0.000250)	0.000897*** (0.000250)	0.000953** (0.000424)	0.00128* (0.000687)
Observations	2031	2031	2031	626	306
Day FE	Yes	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes	Yes

Standard errors in parentheses

* $p < .1$, ** $p < .05$, *** $p < .01$

Building on the analysis in Table H1, section-specific estimates show the strongest contrast in U.S. news: a one-standard-deviation increase in average pro-Democrat score is associated with 1.13 additional NYT homepage hours relative to WSJ. This makes the U.S. section a natural place for an additional robustness check based on common political shocks. Appendix I reports the full section breakdown (Table I1).

Appendix H.1 Matched U.S. Political Shocks

Because the matched-news estimates are strongest in U.S. news, we also re-estimate the homepage-duration regressions using matched political events from the U.S. sections of the NYT and WSJ. The idea is to focus on common political shocks and ask whether relative homepage persistence differs when the underlying event is more favorable to Democrats

than to Republicans. Table H2 reports the results. As in the rest of this appendix, the exercise is informative as a robustness check, but it is not intended as a separate identification design.

Column (1) reproduces the baseline U.S.-section specification using the article-level pro-Democrat score and yields a coefficient of 6.293 hours. Columns (2)-(5) instead classify the direction of the matched event itself. We consider two coding schemes. The first is a narrow Trump/Biden classification that codes events directly tied to Trump, Biden, or their immediate political camps. The second is a broader general political classification that also includes other clearly directional political events, including issue-based stories whose partisan content is less tightly tied to a single politician.

We also focus on two subsamples designed to sharpen the common-shock comparison. In columns (2) and (3), we keep matched pairs that both outlets place prominently on their homepages, which isolates high-salience events where editorial allocation is likely to be most deliberate. In columns (4) and (5), we keep matched pairs in the upper tercile of NYT-WSJ textual similarity, which more tightly restricts the sample to cases where both outlets are covering the same underlying event.

The coefficient on favorable-to-Democrat news is positive in all four shock-based specifications. In the both-prominent sample, the estimate is 1.297 hours under the Trump/Biden classification and 1.148 hours under the broader classification. In the high-similarity sample, the corresponding estimates are 6.585 and 0.657 hours. The larger estimate in the narrow, high-similarity specification is consistent with a simple measurement-error interpretation: the tighter political classification and closer textual match reduce attenuation from noisier event coding. At the same time, the estimates remain imprecise, so the evidence is best read as reinforcing the matched-news appendix rather than as a separate design. The main takeaway is that the relative homepage persistence of pro-Democratic shocks becomes more positive when the matched events are more salient and more clearly shared across the two outlets.

Table H2: OLS Regression of Survival Difference for Matched U.S. Political Shocks (NYT - WSJ)

	(1) Baseline	(2) Both Prominent TB	(3) Both Prominent General	(4) High Similarity TB	(5) High Similarity General
Pro-Dem. Scores (Abstract)	6.293** (2.747)				
Favorable to Dem.		1.297 (4.079)	1.148 (2.061)	6.585 (5.861)	0.657 (2.310)
Tweets Dif. (NYT - WSJ)	0.000650** (0.000313)	-0.000167 (0.000739)	-0.000392 (0.000592)	0.00101 (0.000969)	0.00111 (0.000743)
Observations	546	37	91	40	104
Day FE	Yes	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes	Yes

Standard errors in parentheses

* $p < .1$, ** $p < .05$, *** $p < .01$

Appendix I Supplementary Matched-News Tables

This appendix reports the section-level matched-news decomposition that complements Appendix H.

Table I1: OLS Regression of Survival Difference by Sections (NYT - WSJ)

	(1) Business	(2) US	(3) World	(4) All
Pro-Dem. Scores (Abstract)	-1.658 (4.005)	6.293** (2.747)	-1.585 (3.751)	4.791*** (1.653)
Tweets Dif. (NYT - WSJ)	0.00348*** (0.00127)	0.000650** (0.000313)	0.00107* (0.000646)	0.000886*** (0.000250)
Observations	339	546	321	2031
Day FE	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes

Standard errors in parentheses

* $p < .1$, ** $p < .05$, *** $p < .01$

Appendix J Editorial Decisions on Print Edition and Podcasts

This appendix provides complementary evidence on how slant relates to editorial choices across adjacent products. The comparison is motivated by models in which demand elasticity can shape the relationship between slant and editorial emphasis. However, the evidence here is necessarily more indirect than the homepage-duration analysis because print and podcast decisions may involve different editorial teams, different audiences, different production constraints, and different strategic objectives.

[Gentzkow and Shapiro \(2006\)](#), [Gentzkow et al. \(2015\)](#), and [Puglisi and Snyder \(2015\)](#) emphasize that competition can matter for media distortions, while [Mullainathan and Shleifer \(2005\)](#) shows that the comparative statics can reverse under demand-driven bias. These ideas motivate our cross-platform comparisons, but they do not imply that differences across homepage, print, and podcasts can be mapped cleanly into owner-level or firm-level supply preferences. We therefore treat the estimates below as informative heterogeneity across editorial products and platforms, not as decisive identification of supply-side bias.

In the sections below, we consider specifications of the form,

$$\text{logit}(\Pr(Y_i = 1 \mid X_i)) = \gamma X_i, \tag{J1}$$

where Y_i is an indicator of interest (whether article i is ever included in the print edition in Section J.1 or featured in a podcast in Section J.2). X_i includes article-specific variables that proxy for demand, tone, and slant. Because the coefficients are in log-odds units, we focus on signs, statistical significance, and qualitative patterns rather than direct percentage-point interpretations. Any slant relationship in these regressions should therefore be read as conditional correlation rather than a clean decomposition of demand and supply. The corresponding linear probability and probit specifications deliver similar qualitative results, so the substantive conclusions are unchanged.

Appendix J.1 Print Edition

The readership of print editions for major newspapers like the NYT and WSJ exhibits notably inelastic demand characteristics. This inelasticity reflects high loyalty among readers who consistently prefer their chosen publication over alternatives, despite steep price changes or the availability of similar information online.⁴¹

Empirical evidence supports this. According to [Pew Research Center \(2016\)](#), print newspaper readers tend to be older and more affluent, demographics that correlate with higher brand loyalty and less price sensitivity than among younger, more digital-centric audiences. These readers value in-depth analysis and comprehensive reporting, making them resistant to switching to other media even as print subscription costs rise.

[American Press Institute \(2018\)](#) also highlights that print subscribers have strong emotional and historical ties to the newspaper. The report notes that print subscribers are significantly more likely than other subscribers to cite a longstanding relationship or fond memories with the paper as reasons for their loyalty (19% versus 2%). This connection underscores the inelastic nature of their demand.

In examining editorial decisions regarding print-edition articles, we focus on the NYT due to data availability constraints.⁴² We used the presence of a “print headline” from the NYT API as an indicator of an article’s appearance in the print edition. We manually verified the “print headline” from the API against the one listed at the bottom of articles on the NYT website (articles included in the print edition have a note “A version of

⁴¹Between 2000 and 2024 the NYT’s inflation-adjusted print cover price doubled while print subscriptions declined by only a third (this overstates the price response since the NYT significantly reduced home delivery discounts over this period, and many print subscribers migrated to a digital account). The NYT often discusses the loyalty of their print subscribers: in 2010 80% of print customers had subscribed for over two years.

⁴²WSJ articles do not list whether they are in the print edition, unlike the NYT. There is an archive of WSJ print editions available at <https://www.wsj.com/print-edition/today>, but it is only available for the prior three months. We contacted Dow Jones, the WSJ publisher, but they were unable to generate a print archive for our sample period.

this article appears in print on <date> <section> <page>”). This variable accurately represents an article’s inclusion in the print edition up to February 2023. For most of the observation period the API lists 60% to 70% of articles per month as being in the print edition, but starting February 2023, this figure dropped to below 40%. To address this anomaly, we limited our analysis to articles up to January 2023, resulting in a sample size of 63,740 articles.

Table J1 presents section-level logit estimates of print inclusion. Homepage presence and tweet counts positively predict print inclusion. Conditional on these controls, pro-Democrat score is positively associated with print inclusion in several sections, including “US”, “Business”, “Arts”, and “Style”, while “Life” and “World” remain statistically indistinguishable from zero.⁴³

These patterns are informative but not decisive. Print selection may reflect a mix of mechanisms, including different editorial teams or decision-makers, different audience composition across formats, different elasticities of demand, print-specific space and deadline constraints, and other unobserved considerations besides owner-level or firm-level ideological preferences. For that reason, we view the print estimates as complementary evidence on cross-platform editorial allocation rather than as a clean measure of supply-side bias.

⁴³Two sections have non-significant parameters. “Life” groups several smaller Feature print sections (including “dining”, “sports”, “travel”, “magazine”, and “well”), and “World” is likely to face relatively more competition from other news sources.

Table J1: NYT Print Edition Logit Regression by Major Sections

	(1) Business	(2) US	(3) World	(4) Arts	(5) Life	(6) Style
Ever on Homepage	1.552*** (0.0823)	1.980*** (0.0643)	2.548*** (0.0664)	-0.295*** (0.0653)	0.852*** (0.0767)	0.236** (0.114)
ln Tweets	0.367*** (0.0306)	0.441*** (0.0200)	0.382*** (0.0275)	0.163*** (0.0301)	0.122*** (0.0302)	0.285*** (0.0554)
Sentiment Scores (Abstract)	0.282*** (0.0898)	0.303*** (0.0567)	-0.217*** (0.0651)	-0.119* (0.0717)	0.0559 (0.0844)	-0.187 (0.142)
Pro-Dem. Scores (Abstract)	0.335** (0.163)	0.208* (0.110)	-0.0716 (0.150)	1.230*** (0.154)	-0.229 (0.177)	0.610** (0.269)
Observations	5042	9876	7862	8892	5016	2553
Day FE	Yes	Yes	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes	Yes	Yes

Standard errors in parentheses

* $p < .1$, ** $p < .05$, *** $p < .01$

We also report print estimates for each news tone and for the overall sample. Table J2 examines whether an article is included in the print edition. The logit estimates show that articles featured on the homepage or those with a high number of tweets are more likely to be selected for the print edition. The article's pro-Democrat scores remain small and statistically insignificant across tone groups and in the overall sample.

Table J2: NYT Print Edition Logit Regression by News Tone

	(1) News	(2) Opinion	(3) Feature	(4) All
Ever on Homepage	1.653*** (0.0281)	0.938*** (0.158)	0.542*** (0.0356)	1.240*** (0.0217)
ln Tweets Count	0.265*** (0.00892)	0.317*** (0.0222)	0.440*** (0.0128)	0.330*** (0.00689)
Sentiment Scores (Abstract)	0.247*** (0.0286)	-0.167** (0.0687)	0.0655 (0.0405)	0.124*** (0.0218)
Pro-Dem. Scores (Abstract)	0.0460 (0.0559)	0.104 (0.127)	-0.0430 (0.0822)	0.0354 (0.0428)
Observations	38403	5544	22216	66163
Day FE	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes
Tone FE	No	No	No	Yes

Standard errors in parentheses

* $p < .1$, ** $p < .05$, *** $p < .01$

Appendix J.2 Podcasts

Podcasts have rapidly increased in popularity as a news medium. [Pew Research Center \(2023\)](#) finds that about half of Americans listened to a podcast in 2022, with many encountering news content. Meredith Kopit Levien, Chief Executive of the *New York Times*, highlighted this shift during the Times’ third-quarter results in 2020. Levien noted the importance of The Daily, the newspaper’s flagship podcast, which has an audience of 4 million daily, nearly twice the paper’s peak circulation ([Turvill, 2020](#)).

Levien also highlighted the demographic shift in this audience, with most listeners under 50 and many under 40. The podcasts cultivate greater loyalty among this younger demographic who develop “an affinity for the *New York Times*.” This change is important for the paper, since younger readers disproportionately use news aggregators like Google News or Facebook News Feed. [Reuters Institute \(2023\)](#) notes that younger users are less likely to visit a news site directly and more likely to use social media or other intermediaries, potentially reducing loyalty to any single news source.

Podcasts not only attract a large audience but also play a crucial role in solidifying brand loyalty and boosting newspaper subscriptions. Levien argued that podcasts improve the bottom line and potentially boost subscription growth: “It’s helpful in driving affinity to the brand. Given the improved results since the inception of The Daily four years ago, there’s every reason to believe in its impact.”

We focus on the editorial decisions driving article selection for flagship podcasts. Specifically, we analyze articles featured in The Daily by the NYT and The Journal by the WSJ during our sample period. To gather this data, we visited the URLs of the respective NYT and WSJ podcasts daily and extracted the URLs of the “feature articles” presented in each episode. These articles were then matched to our primary dataset. Only articles featured in a podcast within 30 days of their original publication were included in our analysis.⁴⁴

⁴⁴This constitutes more than 90% of articles that featured in the podcasts in our sample.

Sample statistics in Table J3 highlight key characteristics of podcast-selected articles. Only about one percent of articles are chosen for podcasts, largely due to episode production capacity. Podcasts are primarily based on News articles, with almost no Opinion pieces.

Table J3: Summary of News Articles by Podcast Status

	NYT		WSJ	
	Podcast (No)	Podcast (Yes)	Podcast (No)	Podcast (Yes)
Ever on Homepage	0.603 (0.489)	0.882 (0.323)	0.689 (0.463)	0.941 (0.235)
Tweets Count	151.9 (731.9)	725.4 (2407.8)	52.59 (238.9)	270.3 (505.2)
Sentiment Scores (Abstract)	0.0199 (0.432)	-0.128 (0.471)	0.0227 (0.412)	-0.0579 (0.405)
Pro-Dem. Scores (Abstract)	0.600 (0.211)	0.601 (0.215)	0.533 (0.211)	0.529 (0.217)
tone (News)	0.573 (0.495)	0.944 (0.230)	0.501 (0.500)	0.802 (0.400)
tone (Opinion)	0.0849 (0.279)	0 (0)		
tone (Feature)	0.341 (0.474)	0.0525 (0.223)	0.238 (0.426)	0.176 (0.381)
N	81080	612	20383	224
Mean coefficients; SD in parentheses				

Articles featured in the podcasts are far more popular with readers and receive extra promotion. For both newspapers, tweet counts within the first 12 hours post-publication for these articles are about five times higher than for those not featured, highlighting greater reader engagement with podcast-selected content. Additionally, articles featured in podcasts are more likely to have been displayed on the newspapers’ homepages, with a nearly 30 percentage point higher likelihood compared to non-featured articles. In terms of sentiment, articles chosen for podcasts generally have a more negative tone.

Most importantly for our focus, average pro-Democrat scores are very similar between featured and non-featured articles, consistent with weak slant selection in podcasts.

Building on these data, Table J4 presents logit estimates of podcast inclusion similar to the print specification. Separate results are listed for each news tone as well as overall (there is no Opinion column, since no podcasts are based on an Op-Ed). The conclusions discussed below also hold for major news sections.⁴⁵

The left panel presents results for the NYT and the right panel for the WSJ, with broadly similar patterns across columns. Demand-side popularity, as measured by tweet count, is positively associated with podcast inclusion. For the NYT, homepage presence is also positively associated with podcast inclusion. In contrast, pro-Democrat score is generally near zero and imprecisely estimated.

Table J4: Podcast Logit Regression by News Tones

	NYT			WSJ		
	News	Feature	All	News	Feature	All
Ever on Homepage	0.702*** (0.160)	1.117** (0.560)	0.737*** (0.154)	0.179 (0.304)	0 (.)	0.431 (0.299)
ln Tweets	0.512*** (0.0304)	0.785*** (0.119)	0.527*** (0.0295)	0.662*** (0.0522)	1.119*** (0.124)	0.745*** (0.0471)
Sentiment Scores (Abstract)	-0.292*** (0.0990)	-0.453 (0.422)	-0.301*** (0.0965)	-0.302* (0.181)	0.0655 (0.423)	-0.268 (0.165)
Pro-Dem. Scores (Abstract)	-0.0581 (0.205)	-0.844 (0.910)	-0.103 (0.200)	0.231 (0.358)	-1.182 (0.860)	0.0181 (0.328)
Observations	41241	18448	64198	10306	3278	14961
Day FE	Yes	Yes	Yes	Yes	Yes	Yes
Month FE	Yes	Yes	Yes	Yes	Yes	Yes
Tone FE	No	No	Yes	No	No	Yes

Standard errors in parentheses

* $p < .1$, ** $p < .05$, *** $p < .01$

Relative to print, this weaker slant relationship in podcasts is directionally consistent with the idea that formats aimed at broader or more marginal audiences place more weight on broad appeal. At the same time, podcasts can differ from homepages and print editions for many reasons besides ideology, including host-driven selection, longer production cycles, audio-storytelling constraints, and a younger listener base. We therefore interpret the podcast comparison as suggestive heterogeneity across editorial products rather than

⁴⁵Results are available upon request.

as evidence that the paper cleanly identifies supply-side bias.

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